Comparing Mutual Information in Finite Samples: Wald and Expanded Welch–Satterthwaite Tests

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Abstract

Mutual information (MI) measures how much observing one variable reduces uncertainty about another. This thesis develops and evaluates a test for equal MI between two categorical variables across independent populations. The Normal Wald test uses a normal approximation for this comparison, which can give inaccurate false-positive rates in small samples or when many category combinations are rarely observed.

The proposed Expanded Welch–Satterthwaite test (Expanded Welch) retains Wald's estimated MI difference and standard error but compares the shared statistic with a Student t distribution. To choose its degrees of freedom, we derive a first-order approximation to how much each estimated pointwise-MI variance changes between samples. The calculation tracks an observation's contribution to that estimate, including changes in cell probabilities, their row and column totals, and pointwise MI.

Simulations from known populations assess these approximations and compare false-positive rates and power to detect genuine MI differences, while an online-shopping application illustrates the tests on observed data. Near independence, apparent association from sampling noise can make the degrees-of-freedom approximations inaccurate. Expanded Welch brings false-positive rates closer to 5% in selected small-sample settings, with lower power at the usual rejection thresholds, but also reduces rejection where false-positive rates are already below 5%. Across the full simulation grid at a fixed MI difference of 0.01 nats, the power gap becomes small with larger samples. In sparse settings, bias in the shared MI difference can also produce rejection in the opposite direction to the true difference, which changing the cutoff does not resolve.

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Contents

Abstract i

Acknowledgements ii

1 Introduction 1

1.1 Why the comparison is difficult . . . 1
1.2 Research questions . . . 2
1.3 Contributions to the field . . . 3
1.4 Scope and organisation . . . 3

2 Background and Literature Review 5

2.1 Information-theoretic foundations . . . 5
2.1.1 Entropy and information content . . . 5
2.1.2 Pointwise and average mutual information . . . 6
2.2 Estimating discrete mutual information . . . 7
2.3 Sampling variance of plug-in MI . . . 9
2.4 Approaches to testing MI differences . . . 14
2.5 Previous work and this thesis's contribution . . . 19

3 Overview of the Two-Sample MI Test and Wald Baseline 20

3.1 Problem setup and bias-corrected estimator . . . 20
3.2 Shared standard error and test statistic . . . 22
3.3 Normal Wald test and motivation for Expanded Welch . . . 23

4 Detailed Derivation of the Expanded Welch–Satterthwaite Test for MI 25

4.1 Choosing degrees of freedom for Expanded Welch . . . 25

4.2 How pointwise-MI variance changes with the table . . . 28
4.3 Component degrees of freedom and final test . . . 32
4.4 Worked example and implementation . . . 38
4.5 Behaviour and limitations . . . 40

5 Experimental Design **42**
5.1 Setup: Constructing joint distributions with known MI . . . 42
5.2 RQ1: Checking the variance approximations . . . 45
5.3 RQ2: False-positive rates under equal MI . . . 47
5.4 RQ3: Power at a fixed MI difference . . . 48
5.5 Shared evaluation: Sampling and uncertainty . . . 48

6 Results **50**
6.1 RQ1: Accuracy of variance approximations . . . 50
6.1.1 Mean and sampling-variance predictions . . . 50
6.1.2 How moment errors affect degrees of freedom . . . 53
6.1.3 Behaviour near independence . . . 55
6.2 RQ2: False-positive rates under equal MI . . . 59
6.2.1 Overall false-positive patterns . . . 59
6.2.2 Bias and standard-error accuracy . . . 65
6.3 RQ3: Power at a fixed MI difference . . . 68
6.3.1 Uniform margins . . . 68
6.3.2 Same-skew margins . . . 70
6.3.3 Different-skew margins . . . 72
6.4 Application to online shopping data . . . 73

7 Conclusions **76**
7.1 Contributions and main findings . . . 76
7.1.1 RQ1: Accuracy of variance approximations . . . 76
7.1.2 RQ2: False-positive rates under equal MI . . . 77
7.1.3 RQ3: Power at a fixed MI difference . . . 77
7.2 Limitations . . . 78
7.3 Future work . . . 79

A Near-Independence Calculations **80**

A.1 Fixed-margin behaviour near independence 80

A.2 The independence boundary 82

B Experimental Details **86**

B.1 Copying construction and target selection 86

B.2 Parameter grids and sampling 87

B.3 Validity and sparse-sample measures 88

B.4 Bias, variance and degrees of freedom 89

B.5 Simulation uncertainty 90

B.6 Component moment targets and uncertainty 90

B.7 Application comparison family and Holm adjustment 91

List of Figures

2.1 Standard normal distribution and Student t distributions with 3, 10 and 30 degrees of freedom. 16

6.1 Mean and sampling-variance ratios under different-skew margins. 51
6.2 Component degrees-of-freedom approximations relative to their moment targets. 54
6.3 Component moments and degrees of freedom near independence. 57
6.4 False-positive rates under uniform margins. 61
6.5 False-positive rates under different-skew margins. 63
6.6 Power at a fixed MI difference under uniform margins. 69
6.7 Power at a fixed MI difference under same-skew margins. 71
6.8 Power at a fixed MI difference under different-skew margins. 72

List of Tables

3.1 Variance quantities in the two-sample MI test. 23

4.1 Component degrees of freedom and their inputs. 36
4.2 Operational Expanded Welch–Satterthwaite algorithm. 39

5.1 Row and column margins for each population. 45
5.2 Settings for the approximation checks, broad null comparison and power study. 47
5.3 Performance measures calculated from repeated samples. 49

6.1 Change in distance from 5% rejection across the broad null grid. 59
6.2 Bias and standard-error accuracy at selected equal-MI settings. 65
6.3 Null rejection rates with pointwise 95% intervals. 66
6.4 False-positive rates at stronger common MI. 66
6.5 Corrected MI estimates and uncertainty of their differences. 74
6.6 Two-sample test results for online shopping sessions. 75

B.1 Unadjusted p-values in the six-comparison application family. 92

Chapter 1

Introduction

Mutual information (MI) measures the average reduction in uncertainty about one variable obtained by observing another (Bossomaier et al., 2016). For example, a categorical input feature X may carry different amounts of information about a class label Y in two populations. This thesis develops and evaluates a statistical test for that comparison using two independently collected samples.

Let P and Q be the joint probability distributions of X and Y in the two populations. The null and alternative hypotheses are

$$H_0 : I(P) = I(Q), \qquad H_1 : I(P) \neq I(Q). \tag{1.1}$$

Under H_0, the populations have equal MI even though their joint probability tables can differ.

In practice, MI is estimated from observed counts that vary from sample to sample, so two samples can produce different MI estimates even when the population MI values are equal. A statistical test must account for this sampling variation before an estimated difference can be interpreted as evidence of a population difference.

1.1 Why the comparison is difficult

The Normal Wald test (Wald, 1943; van der Vaart, 1998) divides the estimated MI difference by its standard error and compares the resulting statistic with a standard normal distribution. At a 5% significance level,

the intended false-positive rate is 5% when the population MI values are equal. The normal approximation can be inaccurate in small or sparse samples, causing the test to reject equal MI too often or too rarely.

Both the MI difference and its standard error are estimated from count tables, whose cells record how often each pair of categories occurs. MI depends on the cell probabilities and their row and column totals. When samples are small and rare combinations of categories occur only a few times or not at all, the resulting probability estimates can distort both the MI difference and its estimated uncertainty.

The proposed Expanded Welch–Satterthwaite test (Expanded Welch) retains Wald's estimated MI difference and standard error but uses a Student t distribution instead of the standard normal distribution. Pointwise MI assigns an information value to each cell, and its variance measures how these values vary across population observations. Dividing the estimated pointwise-MI variance by sample size gives each population's contribution to the squared standard error. Choosing the degrees of freedom also requires the sampling mean and variance of that pointwise-MI variance estimate. This calculation must account for the fact that changing a cell probability also changes its row and column totals and the pointwise MI values calculated from them. This thesis derives how these probability changes affect the estimated pointwise-MI variance and uses that calculation to obtain degrees of freedom for Expanded Welch.

The Student t distribution gives Expanded Welch a larger rejection cutoff than Normal Wald, so it rejects less often. The lower rejection rate can reduce false positives but can also reduce statistical power, the probability of rejecting equal MI when the population MI values genuinely differ. The evaluation reports false-positive rates under equal population MI and power at a fixed MI difference of 0.01 nats across the same table sizes, category frequencies and sample sizes. Considering these rates together shows where reduced false positives accompany a loss of detection probability.

1.2 Research questions

The thesis addresses three research questions:

RQ1: How accurately do the derived approximations predict the mean and sampling variance of the estimated pointwise-MI variance, and what degrees of freedom do they imply near and away from independence?

RQ2: How do Expanded Welch's false-positive rates compare with those of Normal Wald under equal population MI?

RQ3: How does Expanded Welch's power compare with Normal Wald's for a fixed population MI difference as sample size, table size and category frequencies change?

1.3 Contributions to the field

1. **Mathematical contribution.** The thesis derives how changing one observation affects the estimated pointwise-MI variance, including the resulting changes in the row and column totals and in pointwise MI.
2. **Methodological contribution.** Expanded Welch uses this variance sensitivity to choose the degrees of freedom for a two-sample test using a Student t distribution.
3. **Empirical contribution.** Controlled simulations check the variance approximations and compare false-positive rates, power and usable results as sample size, table size, category frequencies and MI change.
4. **Practical contribution.** The evaluation identifies selected populations at common MI 0.02 nats where Expanded Welch brings false-positive rates closer to 5% than Normal Wald, quantifies the accompanying detection cost across the fixed-gap power grid, and identifies estimation errors that make both tests unreliable.

1.4 Scope and organisation

This thesis compares MI between two populations using categorical samples summarised in count tables with the same fixed row and column categories. The two samples are independent, and observations within each sample are also independent. For the main derivation, both populations are assumed to have positive MI and a positive probability in every cell. The case of zero MI, where X and Y are independent, is considered separately. The main simulations compare Normal Wald and Expanded Welch using two-sided tests at the 5% significance level.

Chapter 2 reviews the background and literature. Chapters 3 and 4 define the statistic and derive Expanded Welch. Chapters 5 and 6 present the experiment, results and online-shopping application. Chapter 7 brings together the findings, limitations and future work. Appendix A gives supporting near-independence calculations, and Appendix B gives experimental construction, validity and uncertainty details.

Chapter 2

Background and Literature Review

Testing for equal MI in two populations requires an estimate from each sample and its sampling variance. This chapter develops the MI estimator, derives its variance and reviews tests for equal population MI.

2.1 Information-theoretic foundations

2.1.1 Entropy and information content

For a discrete random variable X with probability mass function $p_X(x)$, the information content of outcome x is $-\log p_X(x)$. Rare outcomes have larger information content because they are less expected. Using natural logarithms gives information in nats. The Shannon entropy is the expected information content,

$$H(X) = -\sum_x p_X(x) \log p_X(x). \tag{2.1}$$

Entropy summarises the uncertainty of one random variable before its value is observed (Shannon, 1948).

For a pair (X, Y), the joint entropy $H(X, Y)$ is defined from the joint probabilities in the same way. The conditional entropy $H(X \mid Y) = H(X, Y) - H(Y)$ measures the uncertainty remaining in X after observing Y.

2.1.2 Pointwise and average mutual information

Let $P = (p_{ij})$ be the joint distribution of categorical variables $X \in \{1, \ldots, r\}$ and $Y \in \{1, \ldots, c\}$. The numbers of categories, r and c, are their alphabet sizes. Write

$$p_{i+} = \sum_{j=1}^{c} p_{ij}, \qquad p_{+j} = \sum_{i=1}^{r} p_{ij} \tag{2.2}$$

for the marginal probabilities, or row and column margins. The pointwise mutual information (also called local MI) associated with cell (i, j) is

$$\ell_P(i, j) = \log\left(\frac{p_{ij}}{p_{i+}p_{+j}}\right). \tag{2.3}$$

The denominator is the cell probability under independence with the same margins. Positive pointwise MI means that the cell occurs more often than independence predicts, whereas negative pointwise MI means that it occurs less often.

Mutual information is the probability-weighted average of these pointwise values (Shannon, 1948; Bossomaier et al., 2016):

$$I(P) = \sum_{i=1}^{r}\sum_{j=1}^{c} p_{ij}\ell_P(i, j). \tag{2.4}$$

Here the summations are nested: for each row i, the inner sum runs over all columns j. Equivalently, the expression sums once over every cell in the table and may be written with the shorthand $\sum_{i,j}$.

Because P is the joint distribution of (X, Y), we also write $I(P)$ as $I(X; Y)$. Its entropy form is

$$I(X; Y) = H(X) + H(Y) - H(X, Y). \tag{2.5}$$

MI is therefore the average reduction in uncertainty about X obtained by observing Y, or equivalently about Y obtained by observing X. It is nonnegative and equals zero exactly when $p_{ij} = p_{i+}p_{+j}$ for every cell, including zero-probability cells (Bossomaier et al., 2016).

Although MI measures the overall departure from independence, it does not specify the shape of association. Two populations can have the same MI while assigning probability to different cells, having different margins, or exhibiting different association patterns. Equality of MI is therefore weaker than equality of

distributions, so the null hypothesis requires equal MI without requiring identical cell probabilities in the two population tables.

The logarithm base determines the reporting unit, and this thesis uses natural logarithms to express MI in nats. Dividing by $\log 2$ converts MI to bits. This conversion rescales MI and its standard error together but leaves the degrees of freedom unchanged, so the test result is identical in either unit.

2.2 Estimating discrete mutual information

Suppose a sample of size n produces counts N_{ij}. The empirical cell probabilities are

$$\widehat{p}_{ij} = \frac{N_{ij}}{n}. \tag{2.6}$$

The plug-in estimator inserts these observed relative frequencies into Equation (2.4):

$$\widehat{I}(P) = \sum_{i,j:\widehat{p}_{ij}>0} \widehat{p}_{ij} \log\left(\frac{\widehat{p}_{ij}}{\widehat{p}_{i+}\widehat{p}_{+j}}\right). \tag{2.7}$$

The zero-cell convention follows from $\lim_{x\downarrow 0} x \log x = 0$, so empty cells contribute no term to the sum. When a positive-probability cell is unobserved, the empirical table assigns it zero probability, contributing to sparse-sample estimation error.

Under multinomial sampling, the cell counts vary between samples but must sum to n, so an extra observation in one cell leaves fewer observations for the others. For a fixed set of categories, the plug-in estimate approaches population MI as the sample grows, but its finite-sample bias and sampling variance depend on the complete probability table. Changing the probabilities of rare cells can change both the bias and the sampling variance even when the categories and sample size are unchanged (Paninski, 2003).

Bias is the difference between the average estimate across repeated samples and the population value: $\mathrm{Bias}\{\widehat{I}(P)\} = \mathrm{E}\{\widehat{I}(P)\} - I(P)$. For a fixed $r \times c$ table with positive population cell probabilities, the large-sample expansion has a positive leading term:

$$\mathrm{Bias}\{\widehat{I}(P)\} = \frac{(r-1)(c-1)}{2n} + O(n^{-2}). \tag{2.8}$$

The remainder $O(n^{-2})$ collects terms bounded in magnitude by a constant times n^{-2} as n grows for this fixed population table. The leading term grows with table size and shrinks as the sample grows, although it does not describe every small or sparse sample. The result is closely related to Miller's entropy correction (Miller, 1955), and the same histogram-MI term appears in Moddemeijer (1989). Sparse sampling can produce further error beyond this leading approximation (Paninski, 2003).

When $I(P) = I(Q)$ but $n_P \neq n_Q$, unequal sample sizes give the uncorrected MI difference a leading bias of

$$\frac{(r-1)(c-1)}{2}\left(\frac{1}{n_P} - \frac{1}{n_Q}\right). \tag{2.9}$$

For a 2×2 table with $n_P = 50$ and $n_Q = 100$, Equation (2.9) gives 0.005 nats despite equal population MI values. Chapter 3 corrects this sample-size effect by subtracting $(r-1)(c-1)/(2n)$ from each estimate.

Estimation beyond the leading bias correction.

Data are undersampled when too few observations are available to estimate the joint probabilities reliably, which can leave rare category combinations poorly represented or absent from the sample. Panzeri and Treves (1996) derive analytical bias corrections that account for the categories with nonzero population probability, but these remain large-sample approximations. Paninski (2003) shows that substantial error can remain after standard corrections, including the jackknife, when many categories are poorly sampled. To account for categories missing from the sample, Chao and Shen (2003) adjust entropy using sample coverage, the total population probability of the categories that have been observed. Their adjustment also accounts for the different probabilities of discovering common and rare categories.

Bayesian estimators model the unknown probabilities rather than treating observed proportions as the complete population. The Nemenman–Shafee–Bialek (NSB) estimator combines priors, or probability weights on possible populations, to avoid favouring a narrow range of entropy values before observing the data (Nemenman et al., 2002). More recently, Piga et al. (2024) use the observed counts to estimate a Dirichlet prior's strength, which controls how much the probability estimates are drawn towards a uniform distribution, including for unobserved categories.

Entropy estimators can also be combined to estimate MI through $I(X;Y) = H(X) + H(Y) - H(X,Y)$, as in Equation (2.5). Archer et al. (2013) study this route alongside estimators that model the joint probabilities directly. Combining three separately constructed Bayesian entropy estimates need not

correspond to one joint probability model and can produce negative MI estimates. Nevertheless, these combined estimators generally outperform the authors' fully Bayesian MI estimator in their tested examples.

Hernández and Samengo (2019) tailor MI estimation to a well-sampled, low-entropy variable paired with a variable whose many categories are poorly sampled. Their estimator uses the well-estimated marginal and models the conditional probabilities, rather than estimating three entropies separately. Their results also show that combining Bayesian entropy estimates can retain bias under severe undersampling. When MI estimates from two distinct populations are subtracted, their estimation errors need not cancel. The two-sample problem therefore requires both a small bias in the estimated difference and a standard error that describes its sampling variation.

2.3 Sampling variance of plug-in MI

Two MI estimates can differ even when their population values are equal, because each estimate varies between samples. Testing their difference therefore requires a standard error that accounts for this sampling variation. We first derive the established large-sample approximation for the variance of one plug-in MI estimate:

$$\mathrm{Var}\{\widehat{I}(P)\} \approx \frac{V(P)}{n}. \tag{2.10}$$

Here $V(P)$, defined below, measures how much the contribution of one observation varies across the population. The derivation assumes that the population cell probabilities are positive, so the logarithms and their derivatives are well defined. This first-order variance is established for histogram and discrete MI (Moddemeijer, 1989; Brillinger, 2004). Section 2.4 then explains how this result supports a comparison between two populations.

The argument proceeds in three steps. First, sampling changes the empirical cell probabilities. Second, a first-order Taylor expansion translates those probability errors into an MI error. Third, that error can be written as a sample average, whose variance is straightforward to calculate.

Step 1: From probability errors to MI error.

Using the entropy identity in Equation (2.5), MI can be written directly in terms of the joint and marginal probabilities:

$$I(P) = \sum_{i,j} p_{ij} \log p_{ij} - \sum_{i} p_{i+} \log p_{i+} - \sum_{j} p_{+j} \log p_{+j}. \tag{2.11}$$

Each sum contains the same function, $f(u) = u \log u$. Expanding this function lets us translate probability errors into changes in MI. The derivative needed for its first-order Taylor expansion follows directly from the product rule:

$$f'(u) = \frac{d}{du}\{u \log u\} = \log u + u\frac{1}{u} = \log u + 1. \tag{2.12}$$

The general first-order Taylor formula is

$$f(x) \approx f(a) + f'(a)(x - a), \tag{2.13}$$

where a is the value around which the function is expanded and $x - a$ is the distance from that value. Here the new value is the estimated probability $x = \widehat{u}$, the expansion point is the population probability $a = u$, and $f(t) = t \log t$. Therefore, $x - a = \widehat{u} - u$, which is the sampling error in the estimated probability. Substituting these quantities into the general formula gives

$$\widehat{u} \log \widehat{u} \approx \underbrace{u \log u}_{\text{population value}} + \underbrace{(\log u + 1)(\widehat{u} - u)}_{\text{first-order change}}. \tag{2.14}$$

Subtracting the population value from both sides makes the change explicit:

$$\widehat{u} \log \widehat{u} - u \log u \approx (\log u + 1)(\widehat{u} - u). \tag{2.15}$$

Applying this expansion to the three sums in Equation (2.11) gives

$$\begin{aligned} \widehat{I}(P) - I(P) \approx & \sum_{i,j} (\log p_{ij} + 1)(\widehat{p}_{ij} - p_{ij}) \\ & - \sum_{i} (\log p_{i+} + 1)(\widehat{p}_{i+} - p_{i+}) \\ & - \sum_{j} (\log p_{+j} + 1)(\widehat{p}_{+j} - p_{+j}). \end{aligned} \tag{2.16}$$

The margin errors are sums of the corresponding cell errors:

$$\widehat{p}_{i+} - p_{i+} = \sum_j (\widehat{p}_{ij} - p_{ij}), \qquad \widehat{p}_{+j} - p_{+j} = \sum_i (\widehat{p}_{ij} - p_{ij}). \tag{2.17}$$

For example, the row-margin sum can now be written as

$$\begin{aligned} \sum_i (\log p_{i+} + 1)(\widehat{p}_{i+} - p_{i+}) &= \sum_i \left[(\log p_{i+} + 1) \left\{ \sum_j (\widehat{p}_{ij} - p_{ij}) \right\} \right] \\ &= \sum_{i,j} (\log p_{i+} + 1)(\widehat{p}_{ij} - p_{ij}). \end{aligned} \tag{2.18}$$

The inner sum over j is the probability error for a fixed row i. It is multiplied by that row's coefficient $\log p_{i+} + 1$; the outer sum then adds these products over all rows. The column-margin sum can be rearranged in the same way.

After this rearrangement, the same cell error $\widehat{p}_{ij} - p_{ij}$ appears in the joint term, its row-margin term and its column-margin term. Its combined coefficient is

$$\begin{aligned} &(\log p_{ij} + 1) - (\log p_{i+} + 1) - (\log p_{+j} + 1) \\ &\quad = \log \left(\frac{p_{ij}}{p_{i+} p_{+j}} \right) - 1 \\ &\quad = \ell_P(i,j) - 1. \end{aligned} \tag{2.19}$$

Therefore,

$$\widehat{I}(P) - I(P) \approx \sum_{i,j} \{\ell_P(i,j) - 1\} (\widehat{p}_{ij} - p_{ij}). \tag{2.20}$$

The empirical and population cell probabilities both sum to one, so

$$\sum_{i,j} (\widehat{p}_{ij} - p_{ij}) = \sum_{i,j} \widehat{p}_{ij} - \sum_{i,j} p_{ij} = 1 - 1 = 0. \tag{2.21}$$

The constant -1 in Equation (2.19) therefore contributes zero. Explicitly,

$$\begin{aligned}\widehat{I}(P) - I(P) &\approx \sum_{i,j} \{\ell_P(i,j) - 1\}(\widehat{p}_{ij} - p_{ij}) \\ &= \sum_{i,j} \ell_P(i,j)(\widehat{p}_{ij} - p_{ij}) - \sum_{i,j} (\widehat{p}_{ij} - p_{ij}) \\ &= \sum_{i,j} \ell_P(i,j)(\widehat{p}_{ij} - p_{ij}). \end{aligned} \tag{2.22}$$

The complicated changes in the joint table and its two margins have therefore reduced to a weighted sum of cell-probability errors. The weight for a cell is its population pointwise MI.

Step 2: From cell errors to observation-level errors.

Let $Z_a = (X_a, Y_a)$ be observation a. An empirical cell probability is the proportion of observations in that cell, so

$$\widehat{p}_{ij} - p_{ij} = \frac{1}{n} \sum_{a=1}^{n} \{\mathbf{1}(Z_a = (i,j)) - p_{ij}\}. \tag{2.23}$$

Substituting this expression into Equation (2.22) first gives

$$\widehat{I}(P) - I(P) \approx \sum_{i,j} \ell_P(i,j) \left[\frac{1}{n} \sum_{a=1}^{n} \{\mathbf{1}(Z_a = (i,j)) - p_{ij}\} \right]. \tag{2.24}$$

The finite sums can be exchanged, so the contribution from observation a is gathered inside one pair of brackets:

$$\begin{aligned}\widehat{I}(P) - I(P) &\approx \frac{1}{n} \sum_{a=1}^{n} \left[\sum_{i,j} \ell_P(i,j) \mathbf{1}(Z_a = (i,j)) - \sum_{i,j} p_{ij} \ell_P(i,j) \right] \\ &= \frac{1}{n} \sum_{a=1}^{n} \{\ell_P(Z_a) - I(P)\}. \end{aligned} \tag{2.25}$$

The first sum inside the brackets selects the single cell containing Z_a, giving $\ell_P(Z_a)$. The second is the probability-weighted average of all pointwise-MI values, which is $I(P)$.

To first order, Equation (2.25) expresses the MI error as an average of n observation-level contributions. The contribution of an observation in cell (i,j) is simply that cell's pointwise MI minus the population

average, $\ell_P(i,j) - I(P)$. Its mean is zero because

$$\begin{aligned} \mathrm{E}_P\{\ell_P(X,Y) - I(P)\} &= \sum_{i,j} p_{ij}\{\ell_P(i,j) - I(P)\} \\ &= \underbrace{\sum_{i,j} p_{ij}\ell_P(i,j)}_{I(P)} - I(P)\underbrace{\sum_{i,j} p_{ij}}_{1} = 0. \end{aligned} \tag{2.26}$$

The linear approximation therefore has mean-zero error. The higher-order terms omitted by this approximation generate the leading bias described in Section 2.2. The centred function $\ell_P(i,j) - I(P)$ is called the influence function of MI. Equation (2.25) uses it to approximate the sampling error (Hampel, 1974; van der Vaart, 1998; Kandasamy et al., 2015).

Step 3: Calculate the sampling variance.

Define the population variance of these centred contributions by

$$V(P) = \mathrm{Var}_P\{\ell_P(X,Y) - I(P)\} = \sum_{i,j} p_{ij}\{\ell_P(i,j) - I(P)\}^2. \tag{2.27}$$

Under independent sampling, the n terms in Equation (2.25) are independent and each has variance $V(P)$. The covariance between any two different observations is therefore zero. Hence

$$\begin{aligned} \mathrm{Var}\{\widehat{I}(P)\} &\approx \mathrm{Var}\left\{\frac{1}{n}\sum_{a=1}^{n}\{\ell_P(Z_a) - I(P)\}\right\} \\ &= \frac{1}{n^2}\sum_{a=1}^{n}\mathrm{Var}\{\ell_P(Z_a) - I(P)\} \\ &= \frac{1}{n^2}\sum_{a=1}^{n} V(P) = \frac{nV(P)}{n^2} = \frac{V(P)}{n}. \end{aligned} \tag{2.28}$$

Averaging n independent contributions therefore divides their variance by n. For dependent observations, covariance terms must also be included in Equation (2.28), as studied by Moddemeijer (1999). The present experiment uses independent observations.

When $V(P) > 0$, the central limit theorem also gives

$$\frac{\widehat{I}(P) - I(P)}{\sqrt{V(P)/n}} \xrightarrow{d} N(0,1). \tag{2.29}$$

This gives a large-sample normal approximation for a fixed table with positive population cell probabilities. The $1/n$ leading bias from Section 2.2 decreases faster than the $1/\sqrt{n}$ sampling error, so it does not change this limiting distribution.

The independence boundary.

At exact independence, $p_{ij} = p_{i+}p_{+j}$, so $\ell_P(i,j) = \log 1 = 0$ in every cell. Hence $I(P) = 0, V(P) = 0$, and the first-order average in Equation (2.25) vanishes. A finite sample can still have positive MI. Its leading variation comes from the squared cell-probability errors in a second-order Taylor expansion, which produces the usual chi-squared limiting distribution (Brillinger, 2004; Marinescu and Balcau, 2025). Testing independence therefore uses a chi-squared distribution rather than the normal approximation for comparing positive MI values.

2.4 Approaches to testing MI differences

For two independent samples, the variance contributions described in the preceding section add to approximate the sampling variance of the MI difference. Estimating this sum and taking its square root gives the standard error, which estimates the difference's standard deviation across repeated samples. This section reviews the Normal Wald test, the Welch–Satterthwaite principle and other approaches to that comparison. The hypothesis is that the populations have the same MI, $I(P) = I(Q)$, even when their cell probabilities differ.

The Normal Wald test.

A Wald test estimates a parameter difference, divides it by an estimated standard error, and compares the result with a standard normal distribution (Wald, 1943; van der Vaart, 1998). For MI estimated from independent samples from P and Q, the first-order variance contributions add:

$$\operatorname{Var}\{\widehat{I}(P) - \widehat{I}(Q)\} \approx \frac{V(P)}{n_P} + \frac{V(Q)}{n_Q}. \tag{2.30}$$

Each term approximates the sampling variance of one MI estimate, with independence of the two samples making their covariance zero. Replacing the population variances by estimates gives the standardisation

$$\frac{\widehat{I}(P) - \widehat{I}(Q)}{\sqrt{\widehat{V}(P)/n_P + \widehat{V}(Q)/n_Q}} \xrightarrow{d} N(0,1) \qquad \text{under } I(P) = I(Q). \tag{2.31}$$

Here $\widehat{V}(P)$ and $\widehat{V}(Q)$ estimate the pointwise-MI variances $V(P)$ and $V(Q)$. The normal limit assumes positive population cell probabilities, positive pointwise-MI variances and increasing sample sizes. With smaller samples, the MI difference may not follow the normal approximation closely, and the estimated standard error can vary between samples. The Wald test still uses the same standard normal distribution cutoff, regardless of how much the estimated standard error varies. Chapter 3 applies the leading bias correction to the numerator and specifies the full Normal Wald test.

The Welch–Satterthwaite principle.

This thesis adapts the Welch–Satterthwaite principle from comparing population means to comparing population MI values. Welch's test allows the population variances to differ and compares the standardised difference between two sample means with a Student t distribution with ν degrees of freedom (Welch, 1947). Here $\nu > 0$ controls the distribution's tail weight; an effective value need not be an integer. For two independent samples with sample means $\bar{X}_P, \bar{X}_Q$ and sample variances s_P^2, s_Q^2, the statistic for testing equal population means and the effective degrees of freedom are

$$T_{\text{Welch}} = \frac{\bar{X}_P - \bar{X}_Q}{\sqrt{s_P^2/n_P + s_Q^2/n_Q}}, \tag{2.32}$$

$$\nu = \frac{\left(s_P^2/n_P + s_Q^2/n_Q\right)^2}{\dfrac{(s_P^2/n_P)^2}{n_P - 1} + \dfrac{(s_Q^2/n_Q)^2}{n_Q - 1}}. \tag{2.33}$$

The denominator contains one estimated variance contribution from each sample, with component degrees of freedom $n_P - 1$ and $n_Q - 1$ under normal sampling. If s_P^2/n_P dominates, the formula makes the combined ν close to $n_P - 1$; when both contributions matter, both samples' degrees of freedom affect ν. Satterthwaite obtains this form by matching the mean and variance of their sum to a scaled chi-squared variable (Satterthwaite, 1946).

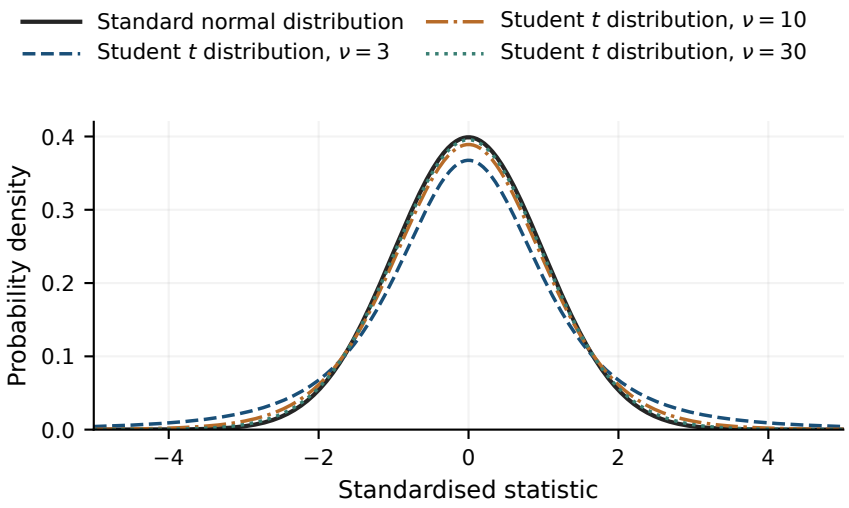


Figure 2.1: Standard normal distribution and Student t distributions with 3, 10 and 30 degrees of freedom.

Figure 2.1 shows that the Student t distribution has heavier tails than the standard normal distribution, especially when ν is small. More probability in the tails means a larger absolute statistic is needed to reject the null at the same significance level. As ν increases, the Student t distribution approaches the standard normal distribution, and their rejection cutoffs become closer.

For MI, the estimated variance contributions are $\widehat{V}(P)/n_P$ and $\widehat{V}(Q)/n_Q$. The ordinary $n_P - 1$ and $n_Q - 1$ values follow from the scaled chi-squared distribution of a sample variance under normal sampling; they do not automatically apply to an estimated pointwise-MI variance. Chapter 4 therefore derives the component degrees of freedom from the sampling variability of $\widehat{V}(P)$ and $\widehat{V}(Q)$, using the same moment-matching principle.

The estimated MI sampling variance $\widehat{V}(P)/n_P$ describes uncertainty in $\widehat{I}(P)$, whereas $\mathrm{Var}\{\widehat{V}(P)\}$ measures how much the variance estimate itself changes between samples. Even variance estimators with the same mean can have different sampling variances, so Satterthwaite's calculation matches both moments. Greater sampling variance relative to the squared mean gives fewer degrees of freedom and a larger rejection cutoff (Satterthwaite, 1946). A sample-size assignment alone does not adapt to differences in this relative variability between probability tables.

The same principle has been used in regression to account for uncertainty in estimated variances (Pan and Wall, 2002; Bell and McCaffrey, 2002; Kauermann and Carroll, 2001; Fay and Graubard, 2001;

Imbens and Kolesár, 2016). For example, Pan and Wall (2002) calculate approximate degrees of freedom from the sampling variability of a regression variance estimate, while Expanded Welch uses the sampling variability of the pointwise-MI variance estimate. These regression precedents address variance estimates that are too small or confidence intervals that cover too rarely (Bell and McCaffrey, 2002; Kauermann and Carroll, 2001). A larger cutoff from the Student t distribution can widen the intervals and reduce excessive rejection in such settings.

Previous work on MI uncertainty and comparisons.

Roulston (1999) derives entropy and MI error estimates using error propagation and examines them numerically. These calculations describe the uncertainty of an individual estimate, a necessary input to a comparison of two estimates. The first-order results of Moddemeijer (1989) and Brillinger (2004) give this uncertainty as $V(P)/n_P$, with $V(P)$ determined by the population table. Replacing its probabilities by observed proportions produces the variance estimate used by Wald to calculate the standard error of the MI difference. Choosing a Welch–Satterthwaite cutoff additionally requires the sampling variance of that variance estimate, which describes how much it changes when a new sample produces a different table.

In a Bayesian approach to uncertainty about population MI, Hutter (2002) and Hutter and Zaffalon (2005) model the unknown cell probabilities and calculate the variance and shape of the resulting MI distribution conditional on the observed table. Our repeated-sampling comparison instead asks how often Normal Wald and Expanded Welch reject equal MI when new samples are drawn from fixed populations P and Q.

Hutcheson's test compares two estimates of Shannon diversity, or entropy, using a Welch-type statistic and a Student t distribution with effective degrees of freedom

$$\nu_H = \frac{(s^2_{H,P} + s^2_{H,Q})^2}{(s^2_{H,P})^2/n_P + (s^2_{H,Q})^2/n_Q} \tag{2.34}$$

(Hutcheson, 1970). Here each of $s^2_{H,P}$ and $s^2_{H,Q}$ is the estimated sampling variance of the corresponding entropy estimator $\widehat{H}$, and each variance component is assigned its sample size, not $n_i - 1$. Expanded Welch instead calculates its degrees of freedom from how the estimated pointwise-MI variance changes with the table, including the effects of a cell change on its row and column totals.

The entropy comparison establishes that a Student t distribution can be used to compare estimated information quantities, but MI requires more than substituting a different numerator. The row, column and joint entropies in MI are calculated from the same observations, so their estimation errors are connected through the cell counts contributing to all three. The MI sampling-variance formulas in Section 2.3 account for these connected estimation errors. To obtain degrees of freedom, Expanded Welch also tracks how a cell change alters both the probability weights and the pointwise-MI values whose weighted variance is being estimated. Thus Hutcheson supplies a close comparison framework, while the cell and margin derivatives supply the MI-specific inputs.

In a segregation application, Mora and Ruiz-Castillo (2009) derive a normal approximation for comparing positive population MI between populations and time periods, providing a direct precedent for the Normal Wald baseline. Their later peer-reviewed account places the MI index within the segregation literature (Mora and Ruiz-Castillo, 2011).

By contrast, Guha and Chothia (2014) test whether two continuous distributions are identical using MI between pooled observations and their sample labels. For discrete data, Drake and Guha (2014) develop an MI-based k-sample test for equality of distributions. These tests compare complete distributions, not the MI of two variables within each population.

Resampling alternatives.

A bootstrap approximates sampling variation by repeatedly drawing samples with replacement from the observed data (Efron, 1979). For this comparison, samples would be drawn separately from the two observed tables, using their original sample sizes, and the MI difference recalculated for each pair. Resampling from separate empirical probability tables allows their cell probabilities to differ and provides an estimated sampling distribution of the MI difference. The bootstrap variation around the observed estimate could then be used to construct an uncertainty interval for the population difference and assess whether that interval includes zero.

Because an ordinary empirical bootstrap assigns an unobserved cell zero probability, it cannot sample that cell and can therefore miss error arising from unobserved cells. Paninski (2003) gives a broader warning: confidence intervals based on resampling can be too narrow when information estimates remain biased.

A simple permutation test combines the two samples and shuffles their group labels, giving an exact test when the populations have the same complete distribution, $P = Q$. Our hypothesis requires only

$I(P) = I(Q)$, allowing the tables to differ while their MI values are equal. Equal MI alone therefore does not guarantee that this label-permutation test has the stated false-positive rate (Chung and Romano, 2013). Although specialised permutation methods allow distinct populations in some parameter comparisons (Janssen, 1997; Chung and Romano, 2013), their validity for equal MI would need to be established.

2.5 Previous work and this thesis's contribution

Existing work provides the leading bias and sampling variance of MI (Miller, 1955; Moddemeijer, 1989; Brillinger, 2004; Paninski, 2003). Hutcheson's entropy comparison assigns degrees of freedom from sample sizes (Hutcheson, 1970), while Pan and Wall (2002) use the variability of a regression variance estimate to choose degrees of freedom. Applying this approach to MI requires calculating how the estimated pointwise-MI variance changes with the observed table, including changes to the cell probabilities, row and column totals, and pointwise MI values.

This thesis derives how changes in the observed count table affect the estimated pointwise-MI variance and uses that result to calculate MI-specific degrees of freedom for Expanded Welch. The resulting test retains Normal Wald's MI difference and standard error but compares the statistic with a Student t distribution rather than a standard normal distribution.

Paninski (2003) shows that finite-sample information estimates can retain substantial bias after a leading correction. Such bias or an inaccurate first-order variance approximation can undermine both tests, even when Expanded Welch lowers rejection. The experiment therefore measures errors in the shared MI difference and standard error alongside false-positive rates and power.

Chapter 3

Overview of the Two-Sample MI Test and Wald Baseline

This chapter defines the estimated MI difference, standard error and test statistic shared by Normal Wald and Expanded Welch, then calculates the Normal Wald p-value using the standard normal distribution.

3.1 Problem setup and bias-corrected estimator

Let

$$N^P = (N_{ij}^P), \qquad N^Q = (N_{ij}^Q) \tag{3.1}$$

be independent $r \times c$ count tables with totals n_P and n_Q. The rows and columns represent the same categorical variables X and Y in both populations, with the same category definitions. The empirical probabilities are

$$\widehat{p}_{ij} = \frac{N_{ij}^P}{n_P}, \qquad \widehat{q}_{ij} = \frac{N_{ij}^Q}{n_Q}. \tag{3.2}$$

The population parameter is the signed difference

$$\Delta = I(P) - I(Q). \tag{3.3}$$

The principal test is two-sided:

$$H_0 : \Delta = 0 \quad \text{against} \quad H_1 : \Delta \neq 0. \tag{3.4}$$

The large-sample argument uses the following assumptions.

Assumption 1 (Sampling)**.** *Observations are independent and identically distributed within each population, and the two population samples are independent.*

Assumption 2 (Aligned fixed alphabets)**.** *The row and column categories are fixed in advance and aligned between the two tables.*

Assumption 3 (Positive cell probabilities)**.** *Every cell in the population table has a probability greater than zero. A sample can still contain no observations in a cell, especially when that cell is rare. We retain such samples and record how often cells have zero observed counts to show how sparse the samples are.*

Assumption 4 (Positive MI)**.** *Both populations have positive MI:* $I(P) > 0$ *and* $I(Q) > 0$*. With the positive cell probabilities in Assumption 3, this is equivalent to requiring positive pointwise-MI variances* $V(P) > 0$ *and* $V(Q) > 0$*. The positive-MI assumption excludes exact independence, where the first-order approximation breaks down.*

Assumption 5 (Logarithm units)**.** *All logarithms in this thesis are natural, so MI differences and standard errors are reported in nats. Changing the logarithm base scales both by the same positive constant and leaves the standardised statistic unchanged. This is a choice of units, not a condition for the test to be valid.*

Bias-corrected estimator.

For the first table, define empirical pointwise MI by

$$\widehat{\ell}_P(i,j) = \log\left(\frac{\widehat{p}_{ij}}{\widehat{p}_{i+}\widehat{p}_{+j}}\right) \qquad \text{when } \widehat{p}_{ij} > 0. \tag{3.5}$$

The plug-in MI estimate is

$$\widehat{I}(P) = \sum_{i,j:\widehat{p}_{ij}>0} \widehat{p}_{ij}\widehat{\ell}_P(i,j). \tag{3.6}$$

The same calculation gives $\widehat{I}(Q)$.

Subtracting the leading fixed-support MI bias in Equation (2.8) (Miller, 1955; Moddemeijer, 1989) from each plug-in estimate gives the bias-corrected MI difference:

$$\widehat{\Delta}_{\mathrm{BC}} = \left\{ \widehat{I}(P) - \frac{(r-1)(c-1)}{2n_P} \right\} - \left\{ \widehat{I}(Q) - \frac{(r-1)(c-1)}{2n_Q} \right\}. \tag{3.7}$$

The two terms in braces are the bias-corrected MI estimates for the separate populations. Their corrections use the declared row and column categories because every population cell has positive probability under the model, even when some cells are unobserved in a sample. We retain negative corrected estimates when calculating the difference. When $n_P = n_Q$, the two corrections cancel, so $\widehat{\Delta}_{\mathrm{BC}}$ is exactly the raw plug-in MI difference. With unequal sample sizes, the difference between the two corrections removes the leading unequal-sample bias described in Section 2.2. A correction that estimates support from occupied cells would be a different estimator.

3.2 Shared standard error and test statistic

Section 2.3 showed that the first-order sampling variance of $\widehat{I}(P)$ is $V(P)/n_P$, where $V(P)$ is the population variance of pointwise MI (Moddemeijer, 1989; Brillinger, 2004). Subtracting the fixed leading bias term in Equation (3.7) does not change this variance.

The observed table estimates the pointwise-MI variance $V(P)$ by

$$\widehat{V}(P) = \sum_{i,j:\widehat{p}_{ij}>0} \widehat{p}_{ij} \{ \widehat{\ell}_P(i,j) - \widehat{I}(P) \}^2. \tag{3.8}$$

Define $\widehat{V}(Q)$ analogously. These estimates describe variation in pointwise MI across observation outcomes; dividing each by its sample size estimates the sampling variance of the corresponding MI estimate.

Independence of the two samples makes the covariance between their MI estimates zero, so we estimate the sampling variance of their difference by adding the two estimated variance contributions:

$$\widehat{\mathrm{SE}}^2 = \frac{\widehat{V}(P)}{n_P} + \frac{\widehat{V}(Q)}{n_Q}. \tag{3.9}$$

Taking the positive square root gives $\widehat{\mathrm{SE}}$, which estimates the standard deviation of $\widehat{\Delta}_{\mathrm{BC}}$ across repeated samples. Table 3.1 distinguishes the variation of pointwise MI across observation outcomes from the sampling variation of the estimates calculated from whole tables.

Table 3.1: Variance quantities in the two-sample MI test.

Quantity	Meaning
$V(P)$	Population variance of pointwise MI; $V(P)/n_P$ approximates $\mathrm{Var}\{\widehat{I}(P)\}$.
$\widehat{V}(P)$	Estimate of $V(P)$ from one sampled count table.
$\widehat{V}(P)/n_P$	Estimated sampling variance of $\widehat{I}(P)$; one contribution to $\widehat{\mathrm{SE}}^2$.
$\mathrm{Var}\{\widehat{V}(P)\}$	Sampling variance of $\widehat{V}(P)$ across repeated samples; used to choose the component degrees of freedom.
$\widehat{\mathrm{SE}}^2$	Estimate of $\mathrm{Var}(\widehat{\Delta}_{\mathrm{BC}})$, calculated as $\widehat{V}(P)/n_P + \widehat{V}(Q)/n_Q$.
$\mathrm{Var}(\widehat{\Delta}_{\mathrm{BC}})$	Sampling variance of $\widehat{\Delta}_{\mathrm{BC}}$; the quantity that $\widehat{\mathrm{SE}}^2$ estimates.

The common standardised statistic is

$$T = \frac{\widehat{\Delta}_{\mathrm{BC}}}{\widehat{\mathrm{SE}}}. \tag{3.10}$$

The sign of T indicates whether P or Q has the larger corrected MI estimate, while its magnitude measures the difference relative to its estimated standard error. Both tests use the same effect estimate and plug-in standard error in Equations (3.7)–(3.10), but compare T with different reference distributions centred at zero under H_0.

3.3 Normal Wald test and motivation for Expanded Welch

When both sample sizes grow while the table dimensions and positive population probabilities stay fixed, the statistic approaches a standard normal distribution under equal population MI (Wald, 1943; van der Vaart, 1998):

$$T \xrightarrow{d} N(0,1) \qquad \text{under } H_0. \tag{3.11}$$

The two-sided Normal Wald p-value is

$$p_{\text{Wald}} = 2\{1 - \Phi(|T|)\}, \tag{3.12}$$

where Φ is the standard normal distribution function. Under this reference distribution for equal MI, the p-value is the probability of a statistic at least as far from zero as the observed T. The two-sided calculation doubles the upper-tail probability $1 - \Phi(|T|)$ because symmetry gives the same probability in the lower tail. At the nominal 5% level, the test rejects when $p_{\text{Wald}} \leq 0.05$.

Why use a Student t distribution?

Wald uses the standard normal distribution at every sample size, even though its standard error is estimated from the observed tables. Expanded Welch uses the sampling variability of $\widehat{V}(P)$ and $\widehat{V}(Q)$ to choose the degrees of freedom of a Student t distribution. Hutcheson's entropy test assigns each variance component n_i degrees of freedom (Hutcheson, 1970), while ordinary Welch's test assigns $n_i - 1$ (Welch, 1947). These assignments do not measure how cell probabilities and margins make the estimated pointwise-MI variance change between samples.

To choose the degrees of freedom, we need the average value of each estimated pointwise-MI variance and how much that estimate varies across samples (Satterthwaite, 1946). For P, to first order,

$$\mathrm{E}\{\widehat{V}(P)\} \approx V(P). \tag{3.13}$$

For the remaining input, Chapter 4 derives a first-order approximation to $\mathrm{Var}\{\widehat{V}(P)\}$ by tracking how the estimated pointwise-MI variance changes when one observation changes the joint table.

Chapter 4

Detailed Derivation of the Expanded Welch–Satterthwaite Test for MI

Expanded Welch keeps the estimated MI difference, standard error and test statistic from Chapter 3, but changes the distribution used to judge that statistic from the standard normal distribution to a Student t distribution. To choose its degrees of freedom, this chapter calculates how much the estimated pointwise-MI variance changes between samples. The name Expanded Welch refers to this expansion of the pointwise-MI variance as the table probabilities change.

4.1 Choosing degrees of freedom for Expanded Welch

The squared standard error contains two estimated variance contributions, $\widehat{V}(P)/n_P$ and $\widehat{V}(Q)/n_Q$. We assign component degrees of freedom to each contribution, then combine them to choose the test's Student t distribution. For component inputs ν_P and ν_Q, the Welch–Satterthwaite equation is (Satterthwaite, 1946; Welch, 1947):

$$\nu = \frac{\left\{\widehat{V}(P)/n_P + \widehat{V}(Q)/n_Q\right\}^2}{\left\{\widehat{V}(P)/n_P\right\}^2/\nu_P + \left\{\widehat{V}(Q)/n_Q\right\}^2/\nu_Q}. \tag{4.1}$$

The derivation has four steps. First, we relate the degrees of freedom to the average and across-sample variation of $\widehat{V}(P)$. Second, we calculate how $V(P)$ changes when a small amount of probability moves into one cell. Third, that change tells us how variable $\widehat{V}(P)$ is. Finally, we combine the degrees of freedom from P and Q using Equation (4.1).

The role of the variance estimate follows from the standard construction of a variable with a Student t distribution. If Z has a standard normal distribution and U independently has a chi-squared distribution with ν degrees of freedom, the scaled ratio follows a Student t distribution:

$$\frac{Z}{\sqrt{U/\nu}} \sim t_\nu. \tag{4.2}$$

Dividing by the independent scaled chi-squared term gives the heavier tails of the Student t distribution. For MI, we approximate this construction by modelling the estimated variance with a scaled chi-squared variable whose mean and variance match its sampling moments. The exact identity assumes the stated normal distribution and independence; the MI reference is an approximation.

Moment matching for one pointwise-MI variance estimate.

To obtain the component target $\nu_{P,\mathrm{mom}}$, Satterthwaite matches the mean and variance of the nonnegative estimate $\widehat{V}(P)$ with those of the scaled chi-squared variable

$$\mathrm{E}\{\widehat{V}(P)\}\frac{\chi^2_{\nu_{P,\mathrm{mom}}}}{\nu_{P,\mathrm{mom}}}. \tag{4.3}$$

For a chi-squared variable with $\nu_{P,\mathrm{mom}}$ degrees of freedom,

$$\mathrm{E}(\chi^2_{\nu_{P,\mathrm{mom}}}) = \nu_{P,\mathrm{mom}}, \qquad \mathrm{Var}(\chi^2_{\nu_{P,\mathrm{mom}}}) = 2\nu_{P,\mathrm{mom}}. \tag{4.4}$$

The scaling in Equation (4.3) therefore matches the mean automatically:

$$\mathrm{E}\left[\mathrm{E}\{\widehat{V}(P)\}\frac{\chi^2_{\nu_{P,\mathrm{mom}}}}{\nu_{P,\mathrm{mom}}}\right] = \mathrm{E}\{\widehat{V}(P)\}. \tag{4.5}$$

The variance of the scaled chi-squared variable is

$$\mathrm{Var}\left[\mathrm{E}\{\widehat{V}(P)\}\frac{\chi^2_{\nu_{P,\mathrm{mom}}}}{\nu_{P,\mathrm{mom}}}\right] = \frac{[\mathrm{E}\{\widehat{V}(P)\}]^2}{\nu^2_{P,\mathrm{mom}}}\,\mathrm{Var}(\chi^2_{\nu_{P,\mathrm{mom}}}) \tag{4.6}$$

$$= \frac{2[\mathrm{E}\{\widehat{V}(P)\}]^2}{\nu_{P,\mathrm{mom}}}. \tag{4.7}$$

We now choose $\nu_{P,\mathrm{mom}}$ so that the scaled chi-squared variable also has the same variance as $\widehat{V}(P)$. Equating its variance, calculated above, to the sampling variance of $\widehat{V}(P)$ requires

$$\frac{2[\mathrm{E}\{\widehat{V}(P)\}]^2}{\nu_{P,\mathrm{mom}}} = \mathrm{Var}\{\widehat{V}(P)\}.$$

This condition determines the degrees of freedom within the scaled chi-squared approximation. Solving for $\nu_{P,\mathrm{mom}}$ gives

$$\boxed{\nu_{P,\mathrm{mom}} = \frac{2[\mathrm{E}\{\widehat{V}(P)\}]^2}{\mathrm{Var}\{\widehat{V}(P)\}}}. \tag{4.8}$$

The squared standard error contains $\widehat{V}(P)/n_P$, so we apply moment matching to this contribution as well as to $\widehat{V}(P)$. Because the sample size n_P is fixed, its reciprocal can be taken outside the expectation and its square outside the variance. Applying the same moment-matching calculation to this variance contribution therefore gives

$$\begin{aligned}\frac{2[\mathrm{E}\{\widehat{V}(P)/n_P\}]^2}{\mathrm{Var}\{\widehat{V}(P)/n_P\}} &= \frac{2[\mathrm{E}\{\widehat{V}(P)\}]^2/n_P^2}{\mathrm{Var}\{\widehat{V}(P)\}/n_P^2} \\ &= \frac{2[\mathrm{E}\{\widehat{V}(P)\}]^2}{\mathrm{Var}\{\widehat{V}(P)\}} = \nu_{P,\mathrm{mom}}.\end{aligned}$$

The factors $1/n_P^2$ cancel, so dividing the estimate by n_P leaves its moment-matched degrees of freedom unchanged.

To evaluate Equation (4.8), we need the mean and variance of $\widehat{V}(P)$ across repeated samples from P. The first-order expansion derived in Section 4.3 gives

$$\mathrm{E}\{\widehat{V}(P)\} \approx V(P).$$

At this order, the change in $\widehat{V}(P)$ is linear in the empirical cell-probability errors. Each empirical cell proportion has expectation equal to its population probability, so these errors and their linear contribution have mean zero. Bias arises from terms beyond this linear approximation.

With the mean approximated by $V(P)$, the remaining task is to calculate $\mathrm{Var}\{\widehat{V}(P)\}$. Because sampling changes the empirical cell probabilities and hence the plug-in variance estimate, we first calculate how $V(P)$ responds to small changes in those probabilities.

4.2 How pointwise-MI variance changes with the table

Fix a cell (x, y) and define a distribution that moves a small amount of probability toward this cell (Hampel, 1974; van der Vaart, 1998):

$$P_\varepsilon = (1-\varepsilon)P + \varepsilon\delta_{(x,y)}, \tag{4.9}$$

where $\delta_{(x,y)}$ places probability one on cell (x, y). At $\varepsilon = 0$, the distribution is P. Increasing ε moves a small amount of probability toward the selected cell while preserving total probability one. This direction represents the table error contributed by an observation in cell (x, y). The empirical table averages such one-observation tables, so averaging their first-order effects will give a first-order approximation to the sampling error of $\widehat{V}(P)$. Section 4.3 shows that averaging step after we calculate the effect for one cell.

For an arbitrary cell (i, j),

$$p_{ij}(\varepsilon) = (1-\varepsilon)p_{ij} + \varepsilon\mathbf{1}\{i = x, j = y\}. \tag{4.10}$$

The joint and marginal probability derivatives at the original distribution are

$$\left.\frac{\mathrm{d}p_{ij}(\varepsilon)}{\mathrm{d}\varepsilon}\right|_0 = \mathbf{1}\{i = x, j = y\} - p_{ij}, \tag{4.11}$$

$$\left.\frac{\mathrm{d}p_{i+}(\varepsilon)}{\mathrm{d}\varepsilon}\right|_0 = \mathbf{1}\{i = x\} - p_{i+}, \tag{4.12}$$

$$\left.\frac{\mathrm{d}p_{+j}(\varepsilon)}{\mathrm{d}\varepsilon}\right|_0 = \mathbf{1}\{j = y\} - p_{+j}. \tag{4.13}$$

The derivatives of the joint probabilities sum to zero because

$$\sum_{i,j}\{\mathbf{1}(i = x, j = y) - p_{ij}\} = 1 - 1 = 0. \tag{4.14}$$

Effect of the perturbation on pointwise MI and MI.

Along the perturbation path, pointwise MI is

$$\ell_{P_\varepsilon}(i,j) = \log p_{ij}(\varepsilon) - \log p_{i+}(\varepsilon) - \log p_{+j}(\varepsilon). \tag{4.15}$$

For each logarithm, the derivative is the probability derivative divided by the probability. Substituting the joint and marginal derivatives gives

$$\begin{aligned}\left.\frac{\mathrm{d}}{\mathrm{d}\varepsilon}\ell_{P_\varepsilon}(i,j)\right|_0 &= \frac{\mathbf{1}\{i=x, j=y\} - p_{ij}}{p_{ij}} - \frac{\mathbf{1}\{i=x\} - p_{i+}}{p_{i+}} - \frac{\mathbf{1}\{j=y\} - p_{+j}}{p_{+j}} \\ &= \frac{\mathbf{1}\{i=x, j=y\}}{p_{ij}} - \frac{\mathbf{1}\{i=x\}}{p_{i+}} - \frac{\mathbf{1}\{j=y\}}{p_{+j}} + 1.\end{aligned} \tag{4.16}$$

The joint term supplies -1, while subtracting the row and column terms supplies $+1$ each. Their sum is the final $+1$.

MI along the path is

$$I(P_\varepsilon) = \sum_{i,j} p_{ij}(\varepsilon)\ell_{P_\varepsilon}(i,j). \tag{4.17}$$

The product rule separates its derivative into the change in probability weights and the change in pointwise-MI values:

$$\begin{aligned}\left.\frac{\mathrm{d}I(P_\varepsilon)}{\mathrm{d}\varepsilon}\right|_0 &= \sum_{i,j}\{\mathbf{1}(i=x, j=y) - p_{ij}\}\ell_P(i,j) \\ &\quad + \sum_{i,j} p_{ij}\left.\frac{\mathrm{d}\ell_{P_\varepsilon}(i,j)}{\mathrm{d}\varepsilon}\right|_0.\end{aligned} \tag{4.18}$$

The first sum is $\ell_P(x,y) - I(P)$. Evaluating the second sum using Equation (4.16), the selected-cell term contributes $p_{xy}/p_{xy} = 1$; the row term contributes $-\sum_j p_{xj}/p_{x+} = -1$; the column term likewise contributes -1; and the constant contributes $\sum_{i,j} p_{ij} = 1$. These four contributions cancel. Hence

$$\boxed{\left.\frac{\mathrm{d}I(P_\varepsilon)}{\mathrm{d}\varepsilon}\right|_0 = \ell_P(x,y) - I(P)}. \tag{4.19}$$

This is the centred pointwise-MI influence function from Section 2.3.

Effect of the perturbation on the pointwise-MI variance.

The target is the first-order change in

$$V(P) = \mathrm{Var}_P\{\ell_P(X,Y)\}. \tag{4.20}$$

Write this variance in second-moment form:

$$V(P) = \mathrm{E}_P\{\ell_P(X,Y)^2\} - I(P)^2. \tag{4.21}$$

Define the observation-level variance sensitivity by

$$g_P(x,y) = \left.\frac{\mathrm{d}V(P_\varepsilon)}{\mathrm{d}\varepsilon}\right|_0. \tag{4.22}$$

Differentiating Equation (4.21) gives

$$g_P(x,y) = \left.\frac{\mathrm{d}}{\mathrm{d}\varepsilon}\,\mathrm{E}_{P_\varepsilon}\{\ell_{P_\varepsilon}(X,Y)^2\}\right|_0 - 2I(P)\left.\frac{\mathrm{d}I(P_\varepsilon)}{\mathrm{d}\varepsilon}\right|_0. \tag{4.23}$$

With the MI derivative established, it remains to differentiate the second moment,

$$\mathrm{E}_{P_\varepsilon}\{\ell_{P_\varepsilon}(X,Y)^2\} = \sum_{i,j} p_{ij}(\varepsilon)\ell_{P_\varepsilon}(i,j)^2. \tag{4.24}$$

Applying the product rule to the probability weights and squared pointwise-MI values gives

$$\begin{aligned}\left.\frac{\mathrm{d}}{\mathrm{d}\varepsilon}\,\mathrm{E}_{P_\varepsilon}\{\ell_{P_\varepsilon}(X,Y)^2\}\right|_0 = &\sum_{i,j}\left.\frac{\mathrm{d}p_{ij}(\varepsilon)}{\mathrm{d}\varepsilon}\right|_0 \ell_P(i,j)^2 \\ &+ 2\sum_{i,j} p_{ij}\ell_P(i,j)\left.\frac{\mathrm{d}\ell_{P_\varepsilon}(i,j)}{\mathrm{d}\varepsilon}\right|_0.\end{aligned} \tag{4.25}$$

The change in the probability weights contributes

$$\ell_P(x,y)^2 - \mathrm{E}_P\{\ell_P(X,Y)^2\}. \tag{4.26}$$

To express the margin changes in the second sum, define the probability-weighted row and column means

$$\mathrm{E}_P\{\ell_P(X,Y) \mid X = x\} = \frac{\sum_j p_{xj}\ell_P(x,j)}{p_{x+}}, \tag{4.27}$$

$$\mathrm{E}_P\{\ell_P(X,Y) \mid Y = y\} = \frac{\sum_i p_{iy}\ell_P(i,y)}{p_{+y}}. \tag{4.28}$$

To evaluate the sum multiplied by 2 in Equation (4.25), substitute the pointwise-MI derivative from Equation (4.16). The sum has four terms. The indicators in the first three select the chosen cell, its row and its column:

$$\sum_{i,j} p_{ij}\ell_P(i,j)\frac{\mathbf{1}\{i=x, j=y\}}{p_{ij}} = \ell_P(x,y),$$

$$\sum_{i,j} p_{ij}\ell_P(i,j)\frac{\mathbf{1}\{i=x\}}{p_{i+}} = \frac{\sum_j p_{xj}\ell_P(x,j)}{p_{x+}},$$

$$\sum_{i,j} p_{ij}\ell_P(i,j)\frac{\mathbf{1}\{j=y\}}{p_{+j}} = \frac{\sum_i p_{iy}\ell_P(i,y)}{p_{+y}},$$

$$\sum_{i,j} p_{ij}\ell_P(i,j) = I(P).$$

The row indicator fixes $i = x$ while leaving the sum over columns, so dividing by p_{x+} gives the conditional mean within row x. The column term works in the same way. The final term comes from the constant $+1$ in the pointwise-MI derivative and averages over the entire table. Keeping the minus signs on the row and column terms gives

$$\begin{aligned}
&\sum_{i,j} p_{ij}\ell_P(i,j)\left.\frac{\mathrm{d}\ell_{P_\varepsilon}(i,j)}{\mathrm{d}\varepsilon}\right|_0 \\
&\quad = \ell_P(x,y) - \frac{\sum_j p_{xj}\ell_P(x,j)}{p_{x+}} - \frac{\sum_i p_{iy}\ell_P(i,y)}{p_{+y}} + I(P) \\
&\quad = \ell_P(x,y) - \mathrm{E}_P\{\ell_P(X,Y) \mid X = x\} - \mathrm{E}_P\{\ell_P(X,Y) \mid Y = y\} + I(P).
\end{aligned} \tag{4.29}$$

Combining Equations (4.19), (4.23), (4.26), and (4.29) gives the variance sensitivity. To see how its squared term arises, combine the weight-change term in Equation (4.26) with the derivative of $-I(P)^2$:

$$\begin{aligned}
&\ell_P(x,y)^2 - \mathrm{E}_P\{\ell_P(X,Y)^2\} - 2I(P)\{\ell_P(x,y) - I(P)\} \\
&\quad = \ell_P(x,y)^2 - \{V(P) + I(P)^2\} - 2I(P)\ell_P(x,y) + 2I(P)^2 \\
&\quad = \ell_P(x,y)^2 - 2I(P)\ell_P(x,y) + I(P)^2 - V(P) \\
&\quad = \{\ell_P(x,y) - I(P)\}^2 - V(P).
\end{aligned}$$

The second line uses $\mathrm{E}_P\{\ell_P(X,Y)^2\} = V(P) + I(P)^2$; the last line completes the square. Adding twice the expression in Equation (4.29) gives

$$\boxed{\begin{aligned}
g_P(x,y) = \{\ell_P(x,y) - I(P)\}^2 - V(P) \\
+ 2\big[\ell_P(x,y) - \mathrm{E}_P\{\ell_P(X,Y) \mid X = x\} \\
- \mathrm{E}_P\{\ell_P(X,Y) \mid Y = y\} + I(P)\big].
\end{aligned}} \tag{4.30}$$

The first line describes the change with the pointwise-MI values held fixed, while the bracket accounts for changes in those values as the cell probabilities and their row and column totals change. The conditional means collect the effects of the changing totals, so both parts are needed to describe how the estimated variance varies across samples.

4.3 Component degrees of freedom and final test

Population component degrees of freedom.

Define the population variability of the observation-level sensitivities by

$$\tau^2(P) = \mathrm{Var}_P\{g_P(X,Y)\}. \tag{4.31}$$

Averaging the two parts of Equation (4.30) shows that the observation-level sensitivities $g_P(x,y)$ have a probability-weighted mean of zero. The first part has mean

$$\mathrm{E}_P\left[\{\ell_P(X,Y) - I(P)\}^2 - V(P)\right] = V(P) - V(P) = 0. \tag{4.32}$$

For the bracket, the row and column conditional means are averaged by weighting each row or column by its probability:

$$\begin{aligned}\mathrm{E}_P\left[\mathrm{E}_P\{\ell_P(X,Y) \mid X\}\right] &= \sum_x p_{x+} \frac{\sum_y p_{xy}\ell_P(x,y)}{p_{x+}} \\ &= \sum_{x,y} p_{xy}\ell_P(x,y) = I(P), \qquad (4.33)\\ \mathrm{E}_P\left[\mathrm{E}_P\{\ell_P(X,Y) \mid Y\}\right] &= \sum_y p_{+y} \frac{\sum_x p_{xy}\ell_P(x,y)}{p_{+y}} \\ &= \sum_{x,y} p_{xy}\ell_P(x,y) = I(P). \qquad (4.34)\end{aligned}$$

Each marginal probability cancels the denominator of its conditional mean, leaving the probability-weighted sum over all cells. The bracket therefore also has mean zero, and the full sensitivity satisfies

$$\mathrm{E}_P\{g_P(X,Y)\} = 0 + 2\{I(P) - I(P) - I(P) + I(P)\} = 0. \qquad (4.35)$$

Thus $\tau^2(P)$ measures variation in the sensitivities around zero. We next use these sensitivities to approximate how much $\widehat{V}(P)$ changes between samples.

Assumption 6 (Regular variance sensitivity)**.** *For the first-order Expanded Welch construction,* $\tau^2(P) > 0$ *and* $\tau^2(Q) > 0$.

An observation in cell $Z_a^{(P)}$ can be represented by a table that places probability one in that cell, denoted $\delta_{Z_a^{(P)}}$. Relative to P, that observation contributes the table change $\delta_{Z_a^{(P)}} - P$. The empirical table averages these one-observation tables, so its difference from P is

$$\widehat{P} - P = \frac{1}{n_P} \sum_{a=1}^{n_P} \left\{ \delta_{Z_a^{(P)}} - P \right\}. \qquad (4.36)$$

The plug-in estimate $\widehat{V}(P)$ evaluates V at the empirical table, so $\widehat{V}(P) = V(\widehat{P})$. We approximate its difference from $V(P)$ by a first-order Taylor expansion. For a differentiable scalar function expanded at zero, the general formula is

$$f(t) \approx f(0) + tf'(0).$$

Here set $f(t) = V\{P + t(\widehat{P} - P)\}$. As the scalar t increases, the table moves along the straight line from the population table P at $t = 0$ to the empirical table $\widehat{P}$ at $t = 1$. Thus V is expanded at P, the

new table is $\widehat{P}$, and the table difference is $\widehat{P} - P$. Since $f(0) = V(P)$ and $f(1) = \widehat{V}(P)$, evaluating the Taylor formula at $t = 1$ and subtracting $f(0)$ gives

$$\widehat{V}(P) - V(P) \approx f'(0) = \frac{\mathrm{d}}{\mathrm{d}t} V\{P + t(\widehat{P} - P)\}\Big|_{t=0}.$$

This approximation retains the terms linear in the cell-probability errors and applies when the empirical table is close to the positive population table.

At fixed P, the first derivative is linear in the direction of the table change (van der Vaart, 1998, Chapter 20). Substituting the empirical-table identity above therefore gives

$$\begin{aligned}
&\frac{\mathrm{d}}{\mathrm{d}t} V\{P + t(\widehat{P} - P)\}\Big|_{t=0} \\
&\quad = \frac{1}{n_P} \sum_{a=1}^{n_P} \frac{\mathrm{d}}{\mathrm{d}t} V\{P + t(\delta_{Z_a^{(P)}} - P)\}\Big|_{t=0}.
\end{aligned}$$

Each term follows the one-observation perturbation $P + t(\delta_{Z_a^{(P)}} - P) = (1-t)P + t\delta_{Z_a^{(P)}}$, so its derivative is $g_P(Z_a^{(P)})$ by Equation (4.22). Combining this derivative identity with the Taylor approximation gives

$$\widehat{V}(P) - V(P) \approx \frac{1}{n_P} \sum_{a=1}^{n_P} g_P(Z_a^{(P)}). \tag{4.37}$$

Taking expectations in Equation (4.37) and using the zero mean of g_P gives

$$\begin{aligned}
\mathrm{E}\{\widehat{V}(P)\} - V(P) &\approx \frac{1}{n_P} \sum_{a=1}^{n_P} \mathrm{E}_P\{g_P(Z_a^{(P)})\} \\
&= 0.
\end{aligned}$$

This establishes the first-order mean approximation used in Section 4.1. The independent observations give independent sensitivity contributions, so their covariance terms vanish in the following first-order sampling-variance approximation:

$$\mathrm{Var}\{\widehat{V}(P)\} \approx \frac{1}{n_P^2} \sum_{a=1}^{n_P} \mathrm{Var}_P\{g_P(Z_a^{(P)})\} \tag{4.38}$$

$$= \frac{\tau^2(P)}{n_P}. \tag{4.39}$$

Substituting the first-order mean and variance approximations into Equation (4.8) gives the population first-order approximation to $\nu_{P,\mathrm{mom}}$, defined by

$$\boxed{\nu_{P,\mathrm{FO}} = \frac{2n_P V(P)^2}{\tau^2(P)}}. \tag{4.40}$$

To interpret Equation (4.40), divide the sampling variance of $\widehat{V}(P)$ by $V(P)^2$, the square of the population pointwise-MI variance:

$$\frac{\mathrm{Var}\{\widehat{V}(P)\}}{V(P)^2} \approx \frac{\tau^2(P)}{n_P V(P)^2}, \tag{4.41}$$

so the component degrees of freedom are approximately twice the reciprocal of this relative variance. When $\widehat{V}(P)$ fluctuates little relative to $V(P)$, the degrees of freedom are large and the Student t distribution is close to the standard normal distribution. Greater relative fluctuation instead gives fewer degrees of freedom and a larger rejection cutoff. Positive $V(P)$ does not guarantee positive $\tau^2(P)$: table changes can have no first-order effect on $V(P)$. In that case, describing the sampling variation of $\widehat{V}(P)$ requires a higher-order approximation.

Table-based component degrees of freedom.

For the observed table, we replace every population quantity in Equation (4.30) by its empirical counterpart and sum over occupied cells, since pointwise MI is undefined at an empirical zero. For each occupied cell,

$$\begin{aligned}\widehat{g}_P(i,j) = {} & \{\widehat{\ell}_P(i,j) - \widehat{I}(P)\}^2 - \widehat{V}(P) \\ & + 2\left[\widehat{\ell}_P(i,j) - \frac{\sum_{j'} \widehat{p}_{ij'}\widehat{\ell}_P(i,j')}{\widehat{p}_{i+}} - \frac{\sum_{i'} \widehat{p}_{i'j}\widehat{\ell}_P(i',j)}{\widehat{p}_{+j}} + \widehat{I}(P)\right].\end{aligned} \tag{4.42}$$

An empty cell has zero weight in these sums, so no logarithm or sensitivity needs to be evaluated there. The empirical sensitivities also have probability-weighted mean zero under exact substitution. Their variance is centred explicitly:

$$\widehat{\tau}^2(P) = \sum_{i,j} \widehat{p}_{ij}\left\{\widehat{g}_P(i,j) - \sum_{k,l} \widehat{p}_{kl}\widehat{g}_P(k,l)\right\}^2. \tag{4.43}$$

The estimated component degree of freedom is

$$\boxed{\widehat{\nu}_P = \frac{2n_P \widehat{V}(P)^2}{\widehat{\tau}^2(P)}}. \tag{4.44}$$

Repeating the same calculation for the second table gives $\widehat{\nu}_Q$.

Table 4.1 distinguishes the sampling-moment target, its population first-order approximation and the sample-based estimate. The target and population approximation are fixed for a population and sample size, whereas $\widehat{\nu}_P$ changes with each sampled table.

Table 4.1: Component degrees of freedom and their inputs.

Quantity	Role	Inputs
$\nu_{P,\mathrm{mom}}$	Sampling-moment target	$\mathrm{E}\{\widehat{V}(P)\}$ and $\mathrm{Var}\{\widehat{V}(P)\}$ at fixed P, n_P
$\nu_{P,\mathrm{FO}}$	Population first-order approximation to the target	Population $V(P)$, $\tau^2(P)$ and n_P
$\widehat{\nu}_P$	Component estimate used by Expanded Welch	$\widehat{V}(P)$, $\widehat{\tau}^2(P)$ and n_P from one sample

Combining the two samples' degrees of freedom.

The combining rule follows by applying moment matching to the full estimated variance,

$$\widehat{\mathrm{SE}}^2 = \frac{\widehat{V}(P)}{n_P} + \frac{\widehat{V}(Q)}{n_Q}.$$

Linearity of expectation gives its sampling mean:

$$\mathrm{E}(\widehat{\mathrm{SE}}^2) = \frac{\mathrm{E}\{\widehat{V}(P)\}}{n_P} + \frac{\mathrm{E}\{\widehat{V}(Q)\}}{n_Q}.$$

The two pointwise-MI variance estimates are computed from independent samples, so their covariance is zero and their contributions to $\mathrm{Var}(\widehat{\mathrm{SE}}^2)$ add. Rearranging Equation (4.8) expresses each component's sampling variance in terms of its sampling mean and moment-matched degrees of freedom. Applying this identity to both components gives

$$\begin{aligned}\mathrm{Var}(\widehat{\mathrm{SE}}^2) &= \frac{\mathrm{Var}\{\widehat{V}(P)\}}{n_P^2} + \frac{\mathrm{Var}\{\widehat{V}(Q)\}}{n_Q^2} \\ &= 2\left\{\frac{[\mathrm{E}\{\widehat{V}(P)\}/n_P]^2}{\nu_{P,\mathrm{mom}}} + \frac{[\mathrm{E}\{\widehat{V}(Q)\}/n_Q]^2}{\nu_{Q,\mathrm{mom}}}\right\}.\end{aligned}$$

The combined moment-matched degrees of freedom are twice the squared mean of $\widehat{\mathrm{SE}}^2$ divided by its variance. Substituting the expressions above and cancelling the common factor 2 gives

$$\frac{2[\mathrm{E}(\widehat{\mathrm{SE}}^2)]^2}{\mathrm{Var}(\widehat{\mathrm{SE}}^2)} = \frac{\left\{\mathrm{E}\{\widehat{V}(P)\}/n_P + \mathrm{E}\{\widehat{V}(Q)\}/n_Q\right\}^2}{[\mathrm{E}\{\widehat{V}(P)\}/n_P]^2/\nu_{P,\mathrm{mom}} + [\mathrm{E}\{\widehat{V}(Q)\}/n_Q]^2/\nu_{Q,\mathrm{mom}}}.$$

This establishes the combining rule in Equation (4.1) for population moment inputs. For the observed tables, we replace the sampling means $\mathrm{E}\{\widehat{V}(P)\}$ and $\mathrm{E}\{\widehat{V}(Q)\}$ by $\widehat{V}(P)$ and $\widehat{V}(Q)$, and the component targets $\nu_{P,\mathrm{mom}}$ and $\nu_{Q,\mathrm{mom}}$ by $\widehat{\nu}_P$ and $\widehat{\nu}_Q$. These substitutions give the estimated combined degrees of freedom:

$$\boxed{\widehat{\nu}_{\mathrm{expanded}} = \frac{\left\{\widehat{V}(P)/n_P + \widehat{V}(Q)/n_Q\right\}^2}{\left\{\widehat{V}(P)/n_P\right\}^2/\widehat{\nu}_P + \left\{\widehat{V}(Q)/n_Q\right\}^2/\widehat{\nu}_Q}}. \tag{4.45}$$

The expanded two-sided p-value is

$$p_{\mathrm{expanded}} = 2\left\{1 - F_t(|T|; \widehat{\nu}_{\mathrm{expanded}})\right\}, \tag{4.46}$$

where $F_t(x;\nu)$ denotes the cumulative distribution function of a Student t distribution with ν degrees of freedom, evaluated at x.

For a fixed finite T, the Student t distribution gives a two-sided p-value at least as large as the standard normal distribution used by Wald. Under the implemented validity rules, every valid Expanded Welch result is also Wald-valid. At a common p-value threshold, every Expanded Welch rejection is therefore also a Wald rejection, including when invalid outputs count as non-rejections in the reported rates. At the nominal threshold, Expanded Welch cannot raise rejection where Wald is already conservative. Appendix B.3 specifies the componentwise validity rule, including the exclusion of a zero variance component.

4.4 Worked example and implementation

Consider the two observed 3×3 tables

$$N^P = \begin{pmatrix} 35 & 10 & 5 \\ 10 & 20 & 5 \\ 5 & 5 & 10 \end{pmatrix}, \qquad N^Q = \begin{pmatrix} 30 & 5 & 5 \\ 5 & 30 & 5 \\ 5 & 5 & 10 \end{pmatrix}. \tag{4.47}$$

Their totals are $n_P = 105$ and $n_Q = 100$. In N^P, cell $(1,1)$ has count 35, and its row and column each total 50. Its empirical pointwise MI is therefore

$$\widehat{\ell}_P(1,1) = \log\left\{\frac{35/105}{(50/105)(50/105)}\right\} = \log(1.47) \simeq 0.38526.$$

This cell contributes $(35/105)(0.38526)$ to the plug-in MI estimate. Summing the contributions from all nine cells gives $\widehat{I}(P) = 0.13693$; the same calculation for N^Q gives $\widehat{I}(Q) = 0.25848$ nats. Since $(3-1)(3-1) = 4$, the corrected difference is

$$\widehat{\Delta}_{\mathrm{BC}} = \left(0.13693 - \frac{4}{2(105)}\right) - \left(0.25848 - \frac{4}{2(100)}\right) = -0.12059. \tag{4.48}$$

The variance calculation weights each squared deviation from the estimated MI by its cell probability. Cell $(1,1)$, for example, contributes $(35/105)(0.38526 - 0.13693)^2 \simeq 0.02056$ to $\widehat{V}(P)$. Summing over all cells gives $\widehat{V}(P) = 0.27189$ and $\widehat{V}(Q) = 0.43371$. Hence

$$\widehat{\mathrm{SE}} = \left(\frac{0.27189}{105} + \frac{0.43371}{100}\right)^{1/2} = 0.08323, \qquad T = -1.44898. \tag{4.49}$$

Normal Wald gives $p = 0.14734$.

For Expanded Welch, the same cell illustrates the sensitivity calculation. The first row's pointwise-MI values are 0.38526, -0.51083 and -0.64436, weighted by its counts 35, 10 and 5. Its conditional mean is

$$\frac{35(0.38526) + 10(-0.51083) + 5(-0.64436)}{50} \simeq 0.10308.$$

Because this count table is symmetric, the first column has the same conditional mean. Substituting these values into Equation (4.42) gives

$$\begin{aligned}\widehat{g}_P(1,1) &\simeq (0.38526 - 0.13693)^2 - 0.27189 \\ &\quad + 2\{0.38526 - 0.10308 - 0.10308 + 0.13693\} \\ &\simeq 0.4218.\end{aligned}$$

The probability-weighted mean of the sensitivities is zero, so this cell contributes $(35/105)(0.4218)^2 \simeq 0.0593$ to $\widehat{\tau}^2(P)$. Repeating the calculation for all cells in both tables gives $\widehat{\tau}^2(P) = 0.93212$ and $\widehat{\tau}^2(Q) = 0.82035$. Substitution into Equation (4.44) gives component degrees of freedom

$$\widehat{\nu}_P = 16.65, \qquad \widehat{\nu}_Q = 45.86. \tag{4.50}$$

Combining them with Equation (4.45) gives $\widehat{\nu}_{\text{expanded}} = 59.03$ and $p = 0.15264$. Neither method rejects at the 5% level.

Operational algorithm and complexity.

Table 4.2 summarises one comparison.

Table 4.2: Operational Expanded Welch–Satterthwaite algorithm.

Step	Calculation
1	Validate the two aligned count tables and their declared alphabet dimensions; independence of samples is a design assumption.
2	Compute empirical probabilities, pointwise MI, plug-in MI, and the leading bias-corrected difference.
3	Compute $\widehat{V}(P)$, $\widehat{V}(Q)$, the shared standard error, and T.
4	Compute the row and column pointwise-MI means needed in $\widehat{g}_P$ and $\widehat{g}_Q$.
5	Compute $\widehat{\tau}^2(P)$, $\widehat{\tau}^2(Q)$, and the two component degrees of freedom.
6	Combine the components using Equation (4.45).
7	Return the p-value from the Student t distribution, effective degrees of freedom, and validity diagnostics.

Each empirical probability, pointwise-MI value, row mean, column mean, and variance contribution is calculated by a fixed number of table scans. The time complexity is therefore $O(rc)$, and the working memory is also $O(rc)$.

4.5 Behaviour and limitations

Near independence.

Consider a population table that approaches independence while its row and column margins stay fixed. At independence, every pointwise-MI value is zero. Since the linear MI error in Equation (2.22) weights each cell-probability error by its population pointwise MI, this linear contribution vanishes. Sampling errors can still create apparent association through products of cell-probability errors, which are retained by a second-order expansion. Near independence, the small linear weights allow these quadratic terms to contribute substantially to the MI estimation error.

The fixed-margin expansion in Appendix A.1 relates MI, the pointwise-MI variance $V(P)$, and the variance of the sensitivities $\tau^2(P)$ under the assumptions stated there. Close to independence along that path, with fixed table dimensions and positive cell probabilities, the expansion gives

$$V(P) \approx 2I(P), \qquad \tau^2(P) \approx 4V(P). \tag{4.51}$$

Substituting these local approximations into the component degrees-of-freedom formula gives

$$\nu_{P,\mathrm{FO}} = \frac{2n_P V(P)^2}{\tau^2(P)} \approx \frac{n_P V(P)}{2} \approx n_P I(P). \tag{4.52}$$

At a fixed sample size, this population first-order value tends to zero as positive population MI approaches zero. When $\widehat{V}(P)$ and $\widehat{\tau}^2(P)$ estimate their population counterparts accurately enough for the component formula to be reliable, $\widehat{\nu}_P$ follows this prediction: smaller MI gives fewer degrees of freedom and a larger cutoff from the Student t distribution.

The empirical table, however, need not be independent even when the population is independent. Random differences between its observed cell proportions and the products of its observed margins produce nonzero pointwise-MI values in occupied cells. Both $\widehat{V}(P)$ and $\widehat{\tau}^2(P)$ are calculated from these values, so sampling-induced association enters the estimated degrees of freedom. At a fixed sample size, these estimated inputs need not track the shrinking population quantities accurately enough for $\widehat{\nu}_P$ to follow $n_P I(P)$. Consequently, the estimated degrees of freedom need not approach zero with population MI. Appendix A.2 retains the quadratic sampling terms and derives their contribution to the estimated degrees of freedom.

Theoretical status and validity conditions.

Equations (4.19) and (4.30) are exact derivatives for positive population tables. Using these derivatives to describe finite sampling errors gives first-order approximations for the MI estimate and the estimated pointwise-MI variance. Moment matching then approximates each variance contribution by a scaled chi-squared variable, as in Equation (4.3). The exact Student t construction in Section 4.1 also requires a standard normal numerator independent of its scaled chi-squared denominator. Here, the corrected MI difference and its estimated standard error come from the same tables, so an observation can change both quantities. Matching the variance moments does not establish the required normality or independence, and the Student t distribution for T therefore remains a finite-sample approximation.

For fixed population tables with positive V and τ^2, increasing the sample sizes under regular conditions increases the estimated degrees of freedom. The Student t distribution then approaches the standard normal distribution used by Wald. This large-sample behaviour holds the populations fixed, whereas the near-independence comparison above reduces population MI at a fixed sample size.

Both tests need a finite corrected MI difference and a positive, finite estimated standard error, while Expanded Welch also needs positive, finite component and combined degrees of freedom. A failed requirement gives an invalid calculation, not evidence for the null. The experiment uses the numerical criteria in Appendix B.3 to evaluate validity alongside calibration.

Chapter 5

Experimental Design

The simulations draw count tables from populations with known MI, allowing us to measure errors in the estimated MI difference and its standard error, and to count false positives when the population MI values are equal. The evaluation addresses the three research questions in Section 1.2 by checking Expanded Welch's variance approximations and comparing its false-positive rates and power with those of Normal Wald. The design follows the simulation framework of Morris et al. (2019), specifying the populations, quantities being estimated, methods and performance measures.

One observation records an (X, Y) outcome. One simulation repetition draws a sample of observations from each population, producing two count tables to which both tests are applied.

5.1 Setup: Constructing joint distributions with known MI

Before drawing samples, we construct population probability tables with specified MI values. Equal MI values allow us to measure false-positive rates, while different MI values allow us to measure power. For each table, we first choose how frequently the categories occur, then change the association between X and Y while keeping those frequencies fixed.

The main simulations use square tables in which both variables have k categories, labelled $1, \ldots, k$, giving $k \times k$ cells in the joint distribution. Let $a = (a_1, \ldots, a_k)$ specify the category frequencies: a_i is

the probability of category i, with $a_i > 0$ and $\sum_i a_i = 1$. Within one population, we use the same vector for both variables, so a specifies the row margins for X and the column margins for Y.

To obtain independent variables, we could draw X using a, then draw Y independently using a. Cell (i, j) would have probability $a_i a_j$, and MI would be zero. We introduce dependence by allowing Y to copy the category drawn for X with probability λ. For each observation, first draw X using a, then choose between two ways of generating Y, independently of the value of X:

- With probability λ, set $Y = X$.
- With probability $1 - \lambda$, draw Y independently using a.

Because copying guarantees a match whereas an independent draw matches X only by chance, increasing λ makes matching categories occur more often.

Both choices preserve the category frequencies of Y: copying inherits the probabilities a used to draw X, while an independent draw of Y uses a directly. Consequently, changing how often we copy changes which categories occur together, while the row and column margins remain fixed. Sample row and column proportions fluctuate around these population probabilities.

Each cell probability is the sum of the contributions from these two choices. The independent choice, made with probability $1 - \lambda$, produces $X = i, Y = j$ with probability $a_i a_j$ and therefore contributes $(1 - \lambda)a_i a_j$. The copying choice contributes only to diagonal cells, where the categories match. Reaching cell (i, i) requires both copying and $X = i$, so its contribution is λa_i. Write $\mathbf{1}_{\{i=j\}}$ for an indicator that equals one when $i = j$ and zero otherwise. Adding the two contributions gives

$$p_{ij}(\lambda) = (1 - \lambda)a_i a_j + \lambda a_i \mathbf{1}_{\{i=j\}}, \qquad 0 \leq \lambda < 1, \tag{5.1}$$

where the restriction $\lambda < 1$ keeps the independent choice possible and therefore gives every cell positive probability. This copying construction is a special case of the mixture model described by Bocci et al. (2010).

At $\lambda = 0$, MI is zero. Increasing λ increases MI, although the copying probability needed to reach a particular MI depends on the category frequencies a. To construct a table with the desired MI, we hold a fixed and search numerically for λ, calculating MI from the population table at each candidate value. Appendix B gives the available MI range, the numerical search and the fixed-margin calculation.

Constructing a 2×2 joint distribution with positive MI.

For population P, take $a = (0.7, 0.3)$, so each variable has probability 0.7 of taking category 1 and probability 0.3 of taking category 2. With a target MI of 0.02 nats, the numerical search gives a copying probability of approximately 0.20347.

Cell $(1, 1)$ requires $X = 1$, which has probability 0.7. Given $X = 1$, copying always gives $Y = 1$, while an independent draw gives $Y = 1$ with probability 0.7. Cell $(1, 2)$ also requires $X = 1$, but copying cannot give $Y = 2$. It therefore requires the independent choice followed by a draw of category 2, which has probability 0.3. Multiplying these conditional probabilities by the probability of $X = 1$ gives

$$p_{11} \simeq 0.7\{0.20347 + (1 - 0.20347)(0.7)\} = 0.53273,$$
$$p_{12} \simeq 0.7(1 - 0.20347)(0.3) = 0.16727.$$

Both off-diagonal cells have probability $(1 - \lambda)(0.7)(0.3)$, so $p_{21} = p_{12}$. The second row totals 0.3, giving $p_{22} \simeq 0.3 - 0.16727 = 0.13273$.

For population Q, using $a = (0.8, 0.2)$ and a target MI of 0.03 nats gives a copying probability of approximately 0.26091. The resulting tables are

$$P \simeq \begin{pmatrix} 0.53273 & 0.16727 \\ 0.16727 & 0.13273 \end{pmatrix}, \qquad Q \simeq \begin{pmatrix} 0.68175 & 0.11825 \\ 0.11825 & 0.08175 \end{pmatrix}. \tag{5.2}$$

These tables have $I(P) = 0.02$ and $I(Q) = 0.03$ nats, so the true signed difference is $\Delta = I(P) - I(Q) = -0.01$ nats. To construct an equal-MI comparison, we instead choose the same target MI for both populations and search for each copying probability separately.

Table sizes and margin regimes.

The experiment repeats this construction for square tables of sizes $2 \times 2, 4 \times 4, 8 \times 8$ and 16×16, containing 4, 16, 64 and 256 cells. Each doubling of the number of categories multiplies the number of cells by four, so a fixed sample must be distributed over more possible cells.

At each table size, we compare three patterns of category frequencies, called margin regimes. Uniform margins assign probability $1/k$ to every category. For uneven margins, the first category has probability ρ, and the remaining probability $1 - \rho$ is shared equally among the other $k - 1$ categories. We write this

vector as

$$m_k(\rho) = \left(\rho, \frac{1-\rho}{k-1}, \ldots, \frac{1-\rho}{k-1}\right). \tag{5.3}$$

For example, assigning probability 0.8 to the first of four categories gives

$$m_4(0.8) = \left(0.8, \frac{0.2}{3}, \frac{0.2}{3}, \frac{0.2}{3}\right).$$

In Table 5.1, the same-skew regime uses the same uneven frequencies in both populations, whereas the different-skew regime uses a first-category probability of 0.7 in P and 0.8 in Q. Thus the row and column margins agree within each population, while the margins can differ between populations.

Table 5.1: Row and column margins for each population.

Regime	P: rows and columns	Q: rows and columns
Uniform	$(1/k, \ldots, 1/k)$	$(1/k, \ldots, 1/k)$
Same skew	$m_k(0.8)$	$m_k(0.8)$
Different skew	$m_k(0.7)$	$m_k(0.8)$

Under equal MI, the uniform and same-skew regimes give identical joint probability tables, $P = Q$: both populations have the same margins and the same copying probability. The different-skew regime instead gives different tables with equal MI because the category frequencies differ. Comparing the regimes therefore measures false positives under both identical tables and distinct tables with equal MI.

5.2 RQ1: Checking the variance approximations

Expanded Welch chooses its component degrees of freedom from $\mathrm{E}\{\widehat{V}(P)\}$ and $\mathrm{Var}\{\widehat{V}(P)\}$, and likewise for Q. To check these quantities, we hold the population table P and sample size n_P fixed, then draw repeated count tables. From each table, we estimate the pointwise-MI variance $V(P)$ by $\widehat{V}(P)$, then divide this estimate by n_P to obtain that population's estimated contribution to the variance of the MI difference. The average of the $\widehat{V}(P)$ values estimates $\mathrm{E}\{\widehat{V}(P)\}$; their variation across draws estimates $\mathrm{Var}\{\widehat{V}(P)\}$.

The observation-level sensitivity derived in Chapter 4 describes how the estimated pointwise-MI variance responds to changes in the table. Let $\tau^2(P)$ be its population variance. The first-order predictions are

$$\mathrm{E}\{\widehat{V}(P)\} \approx V(P), \qquad \mathrm{Var}\{\widehat{V}(P)\} \approx \frac{\tau^2(P)}{n_P}. \tag{5.4}$$

We compare $V(P)$ with the observed mean of $\widehat{V}(P)$, and $\tau^2(P)/n_P$ with its observed sampling variance. To examine the effect of estimating the sensitivity from each sample, we also average the sample-based prediction $\widehat{\tau}^2(P)/n_P$ across the repeated tables and compare this average with the same observed sampling variance.

We next compare the component degrees of freedom with the sampling-moment target from Chapter 4:

$$\nu_{P,\mathrm{mom}} = \frac{2\{\mathrm{E}\,\widehat{V}(P)\}^2}{\mathrm{Var}\{\widehat{V}(P)\}}. \tag{5.5}$$

Using the repeated samples to estimate the mean and variance in this target, we compare it with $\nu_{P,\mathrm{FO}}$, calculated from the population probabilities, and the average of the sampled $\widehat{\nu}_P$ values. Averaging $\widehat{\nu}_P$ need not recover $\nu_{P,\mathrm{mom}}$ because each estimate is a nonlinear ratio of sample-based quantities, whereas the target uses the sampling mean and variance of $\widehat{V}(P)$. For the sampled estimates, the ratio $\mathrm{mean}(\widehat{\nu}_P)/\nu_{P,\mathrm{mom}}$ equals one when the average estimate matches the target; a ratio below one means that the estimated degrees of freedom are lower on average. We make the same checks for Q and report the included fraction whenever a summary uses only positive finite degrees of freedom. Appendix B.6 gives the definitions and algebra.

Table 5.2 specifies two grids: the first varies sample size at common MI 0.02 nats, while the second includes very small positive MI to examine behaviour near independence. In this latter range, the predicted sampling variance of $\widehat{V}(P)$ becomes small and quadratic sampling terms can matter. Both grids use all three margin regimes and equal sample sizes $n_P = n_Q = n$. Each row combines every listed table size, MI value and sample size.

Table 5.2: Settings for the approximation checks, broad null comparison and power study.

Check	Parameter values
Across sample sizes	$k \in \{2, 4, 8, 16\}$, $I(P) = I(Q) = 0.02$ nats. $n \in \{10, 20, 50, 100, 250, 500, 1000, 2500, 10000, 50000\}$.
Near independence	$k = 4$, $n \in \{100, 1000, 10000\}$. $I(P) = I(Q)$ takes the values (nats) $0.0001, 0.0003, 0.001, 0.003, 0.01, 0.02, 0.05$.
Fixed-gap power	$k \in \{2, 4, 8, 16\}$, $I(P) = 0.02$, $I(Q) = 0.03$ nats. The same ten sample sizes as the across-sample-size checks.

5.3 RQ2: False-positive rates under equal MI

To measure false positives, we draw a count table from each of two populations with equal MI in every repetition, so every rejection is a false positive. Normal Wald and Expanded Welch use the same bias-corrected MI difference and estimated standard error from Chapter 3, but compare the resulting statistic with different reference distributions. Normal Wald uses the standard normal distribution, while Expanded Welch uses a Student t distribution with MI-specific degrees of freedom estimated from the two sampled count tables.

All reported simulation comparisons use equal sample sizes, so the two leading MI bias corrections cancel from $\widehat{\Delta}_{\mathrm{BC}}$, leaving the raw plug-in MI difference. Residual biases can still differ between distinct population tables, shifting the estimated difference away from its true value even under equal population MI.

Both tests use a two-sided 5% nominal test. The broad null comparison uses the first grid in Table 5.2: common MI 0.02 nats, four table sizes, three margin regimes and ten sample sizes, giving 120 settings. A false-positive rate above 5% means excessive rejection, while a rate below 5% means conservative rejection. Because a rule that never rejects has zero false positives and zero power, we assess nominal calibration by closeness to 5% and evaluate power separately.

We also report the fraction of valid results. Because invalid calculations contribute no rejection to the all-repetition rate, failures can lower that rate without improving the decisions made on usable samples.

5.4 RQ3: Power at a fixed MI difference

To measure power, we draw samples from populations with different MI, so rejection detects a genuine difference. The power study uses the same four table sizes, three margin regimes and ten sample sizes as the broad null comparison in Section 5.3. At each of these 120 settings, the copying construction gives

$$I(P) = 0.02, \qquad I(Q) = 0.03, \qquad \Delta = I(P) - I(Q) = -0.01 \text{ nats}. \tag{5.6}$$

The population tables stay fixed as sample size increases, so the curves show how more observations affect detection of the same MI difference.

For each setting, we generate 100,000 independent pairs of count tables and apply both tests to each pair using their usual p-value threshold of 0.05. Power is the fraction of all repetitions that reject equal MI; invalid calculations count as non-rejections, and the valid-result fraction is reported separately. The sample sizes and margin profiles are specified in Tables 5.2 and 5.1.

Expanded Welch uses the same statistic as Normal Wald with a larger rejection cutoff, so it cannot have higher power at this common threshold. The comparison measures the reduction in detection probability across the grid. We interpret that reduction alongside the false-positive rates in Section 5.3, because a higher rejection rate under an alternative can accompany excessive rejection under equal MI.

5.5 Shared evaluation: Sampling and uncertainty

At one evaluation setting, each repetition draws a count table from P and an independent count table from Q. Both tests use these same two tables, so their rejection decisions can be compared on the same sampled data. Each setting has 100,000 independent repetitions, which determine the precision of the reported rates; each test uses only n_P and n_Q observations within a repetition. Each distinct setting uses separate random streams for the two populations, and repeated settings use the same repetitions wherever they appear. The fixed-gap power study uses new simulation streams.

Table 5.3 defines the measures for each evaluation setting.

Table 5.3: Performance measures calculated from repeated samples.

Measure	Calculation
Rejection rate	$\dfrac{\text{number of valid rejections}}{\text{number of simulation repetitions}}$
Valid-result fraction	$\dfrac{\text{number of valid outputs}}{\text{number of simulation repetitions}}$
Difference bias	$\mathrm{mean}(\widehat{\Delta}_{\mathrm{BC}}) - \Delta$
Variance ratio	$\dfrac{\mathrm{mean}(\widehat{\mathrm{SE}}^2)}{\text{observed variance of } \widehat{\Delta}_{\mathrm{BC}}}$

Bias uses finite MI differences even when a test is invalid. The variance ratio uses repetitions with both finite $\widehat{\Delta}_{\mathrm{BC}}$ and finite $\widehat{\mathrm{SE}}^2$, with one indicating agreement. Ratios below one mean that the squared standard error understates the observed difference variance on average, while ratios above one indicate overstatement. Two-sided power includes rejections in either direction. Appendix B.4 specifies the inclusion rules and complete calculations.

Rejection rates have Wilson 95% confidence intervals (Wilson, 1927), which describe uncertainty from the finite number of simulation repetitions. Intervals for method differences account for both methods being applied to the same simulated data. The intervals apply pointwise to a rate or comparison at one specified setting. Appendix B gives the interval calculations.

Chapter 6

Results

This chapter checks the approximations used to choose Expanded Welch's degrees of freedom, then compares its false-positive rates and power with Normal Wald. Supporting simulations examine additional settings, and the online-shopping application illustrates interpretation of the test outputs.

Each evaluation setting uses 100,000 independent repetitions, each drawing one count table from P and one independently from Q. Normal Wald and Expanded Welch are both applied to the same two sampled tables in each repetition.

6.1 RQ1: Accuracy of variance approximations

For a fixed population table P and sample size n_P, each repetition produces an estimate $\widehat{V}(P)$ of the pointwise-MI variance $V(P)$. We examine its average value, $\mathrm{E}\{\widehat{V}(P)\}$, and its variation between repetitions, $\mathrm{Var}\{\widehat{V}(P)\}$, because both moments enter the component degrees-of-freedom calculation. Table 3.1 in Chapter 3 distinguishes these quantities from the sampling variance of the MI difference.

6.1.1 Mean and sampling-variance predictions

Equation (5.4) predicts $\mathrm{E}\{\widehat{V}(P)\} \approx V(P)$ and $\mathrm{Var}\{\widehat{V}(P)\} \approx \tau^2(P)/n_P$, where $\tau^2(P)$ measures the variability of the observation-level sensitivities derived in Chapter 4. We compare each prediction with the corresponding moment observed across repetitions, and repeat the checks for Q.

Figure 6.1 shows the different-skew populations at common MI 0.02 nats, with row and column margins $m_k(0.7)$ for P and $m_k(0.8)$ for Q. The top row divides the simulated mean of $\widehat{V}(P)$ by the population benchmark $V(P)$, and likewise for Q. The middle row divides $\tau^2(P)/n_P$ by the observed $\mathrm{Var}\{\widehat{V}(P)\}$. The bottom row uses the average $\widehat{\tau}^2(P)/n_P$ as its numerator and the same observed variance as its denominator. Both variance rows use the same vertical scale, so their differences show the effect of substituting sample proportions. A ratio of one indicates agreement. Shading in the first two rows shows approximate 95% Monte Carlo intervals from 100,000 repetitions. Their half-widths are below 1.3% of the plotted values, making the bands barely visible on this scale.

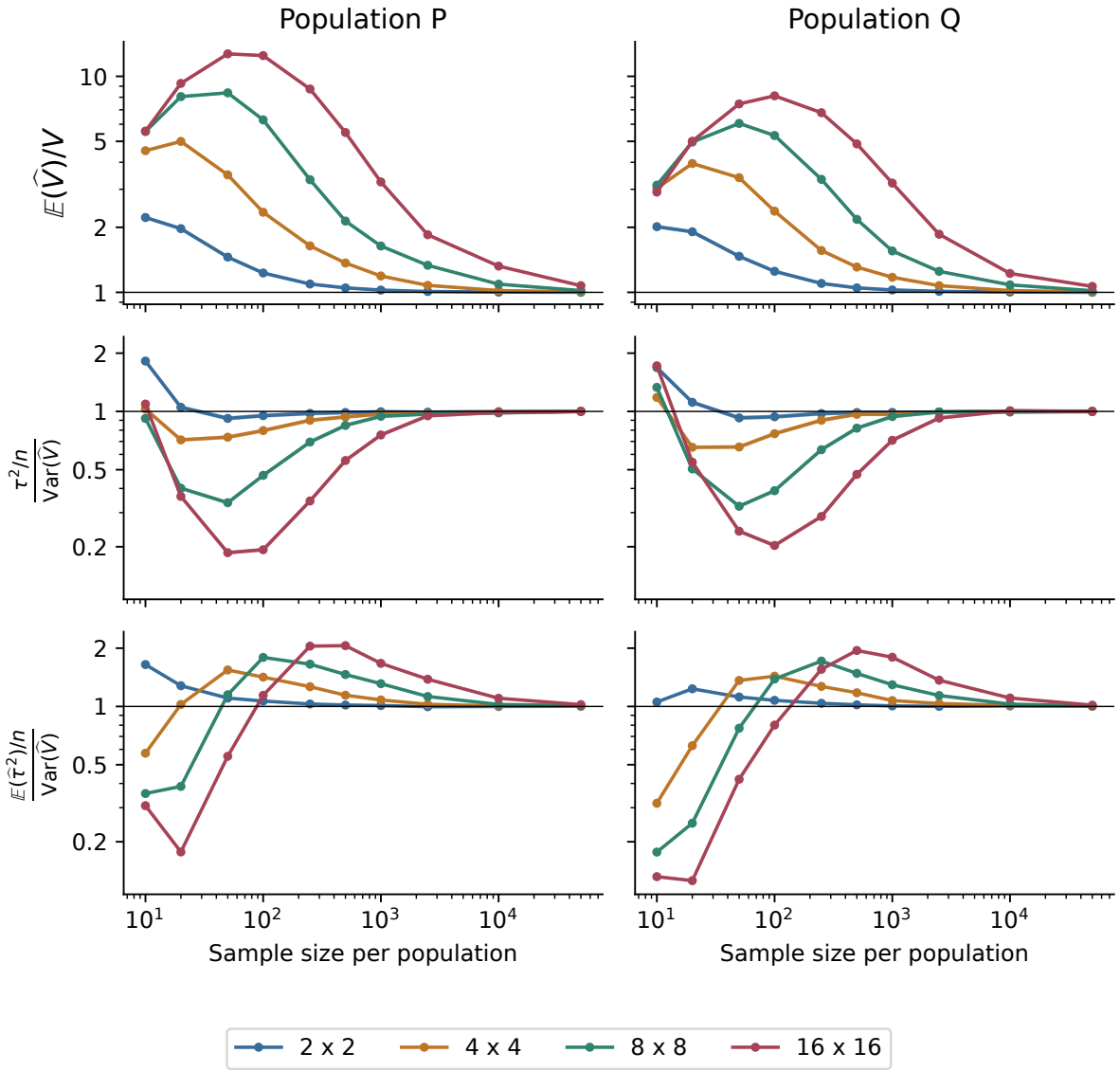


Figure 6.1: Mean and sampling-variance ratios under different-skew margins.

In the top row of Figure 6.1, larger tables have higher peaks in the mean ratio and take more observations to approach one. The 2×2 curve falls towards one without a pronounced peak, whereas the 4×4, 8×8 and 16×16 curves first rise and then fall. The 4×4 peak is smaller and earlier than those for the two larger tables. The rise in these curves occurs as rare combinations that were initially unobserved begin to appear in sampled tables. Cells observed only once can have estimated pointwise MI far from the table's plug-in MI, producing large squared deviations in $\widehat{V}$. For example, if a cell and its row and column each contain one observation, its empirical pointwise MI is $\log n$. Decomposing the cell contributions in 10,000 repetitions at each of seven selected sample sizes from 10 to 10,000 for the 16×16 different-skew populations shows that once-observed cells contribute most of the mean of $\widehat{V}$ around the peak in both populations. Their contribution falls as repeated counts give better estimates of the cell probabilities. Larger tables spread probability over more rare combinations under these margin rules, so this distortion persists to larger sample sizes. Both populations show these patterns, but the larger-table peaks are higher in P than in Q, and the 16×16 peak occurs later in Q. For each of these larger table sizes, the peak mean of $\widehat{V}$ is greater in P than in Q, while $V(P) < V(Q)$. Both the larger numerator and the smaller denominator therefore raise P's peak ratio. Equal MI fixes the population mean of pointwise MI, whereas its variance V also depends on the cell-specific pointwise-MI values and their probability weights, which change with the margins.

In the middle row, the 2×2 curves have shallow troughs, while the larger tables have progressively deeper troughs and take longer to approach one. Ratios below one mean that $\tau^2(P)/n_P$ understates the observed $\mathrm{Var}\{\widehat{V}(P)\}$, with the same interpretation for Q. The dip develops because, for a fixed population, the predicted variance decreases exactly as $1/n$, while the observed sampling variance does not initially follow that decrease. At small sample sizes, many rare combinations remain unobserved. As n increases, some sampled tables contain a given rare cell while others still miss it. An unobserved cell contributes zero to $\widehat{V}$, whereas a single observation can give a large squared deviation of its pointwise MI from the table's plug-in MI. The $1/n$ change in a cell proportion can be large relative to its true probability, so its effect on $\widehat{V}$ can be poorly represented by linearisation around the population table. The observed variance can therefore initially rise, or fall more slowly than $1/n$, while the predicted variance decreases, so the ratio then falls. In the same 16×16 repetitions, subtracting the linear change from $\widehat{V} - V$ leaves a remainder whose variance exceeds that of the linear change around the trough. Including their covariance in the total variance confirms that the linear change alone misses much of the observed variation. With larger samples, repeated counts make the estimated cell probabilities and their log ratios more stable. The variation missed by the linear approximation becomes smaller relative to $\tau^2(P)/n_P$,

and the observed variance approaches this prediction, so the ratio rises back towards one. Under these margin rules, larger tables contain more rare combinations and require more observations before this approximation becomes accurate. The observed variance measures spread about the sampling mean, so it can be predicted accurately while that mean remains above the population benchmark, as in the top row. The same ordering by table size appears in P and Q, but the 16×16 trough occurs later in Q, where the rarer row and column categories have lower probabilities and accumulate observations more slowly.

In the bottom row, the 4×4, 8×8 and 16×16 sample-based predictions first underestimate the observed sampling variance, then overestimate it before approaching agreement. Their ratios begin below one, rise through one to a peak, and then fall towards one, although the 16×16 curves initially dip further before rising. The 4×4 overshoot is smaller and earlier than those for the two larger tables, while 16×16 has the largest overshoot and takes longest to settle near one. The 2×2 curves begin above one and settle near it sooner. Substituting sample proportions changes the sensitivities used to estimate τ^2. Around the overshoot, cells observed only once contribute strongly to $\widehat{\tau}^2$ as well as to $\widehat{V}$, so the averaged sample-based prediction can exceed the observed sampling variance even while the middle-row population prediction remains below it. Sample substitution can therefore reverse the direction of the prediction error. Both populations show this progression, but for the three larger table sizes the initial underprediction is stronger in Q, and the crossing above one occurs later than in P. The subsequent peaks are higher in P than in Q.

6.1.2 How moment errors affect degrees of freedom

We now examine how the moment errors in Figure 6.1 affect the component degrees of freedom for P and Q, using the three quantities and inputs distinguished in Table 4.1. At each setting, the observed mean and sampling variance of $\widehat{V}(P)$ across 100,000 repetitions give the moment target $\nu_{P,\mathrm{mom}}$ in Equation (5.5). The population first-order value $\nu_{P,\mathrm{FO}}$ instead uses $V(P)$ and $\tau^2(P)/n_P$, while the sample-based comparison calculates $\widehat{\nu}_P$ from each sampled table before averaging these estimates.

Figure 6.2 uses the different-skew populations at common MI 0.02 nats, with margins $m_k(0.7)$ for P and $m_k(0.8)$ for Q. Sample size increases along the horizontal axis, and the lines represent table sizes. The top row divides $\nu_{P,\mathrm{FO}}$ by $\nu_{P,\mathrm{mom}}$; the bottom row divides $\mathrm{E}(\widehat{\nu}_P)$ by the same target, and likewise for Q. On the shared vertical scale, one indicates agreement, while values below or above one indicate fewer or

more degrees of freedom than the target. A bottom-row value is shown only when all 100,000 component estimates are positive and finite. Shading in the top row gives approximate 95% Monte Carlo intervals.

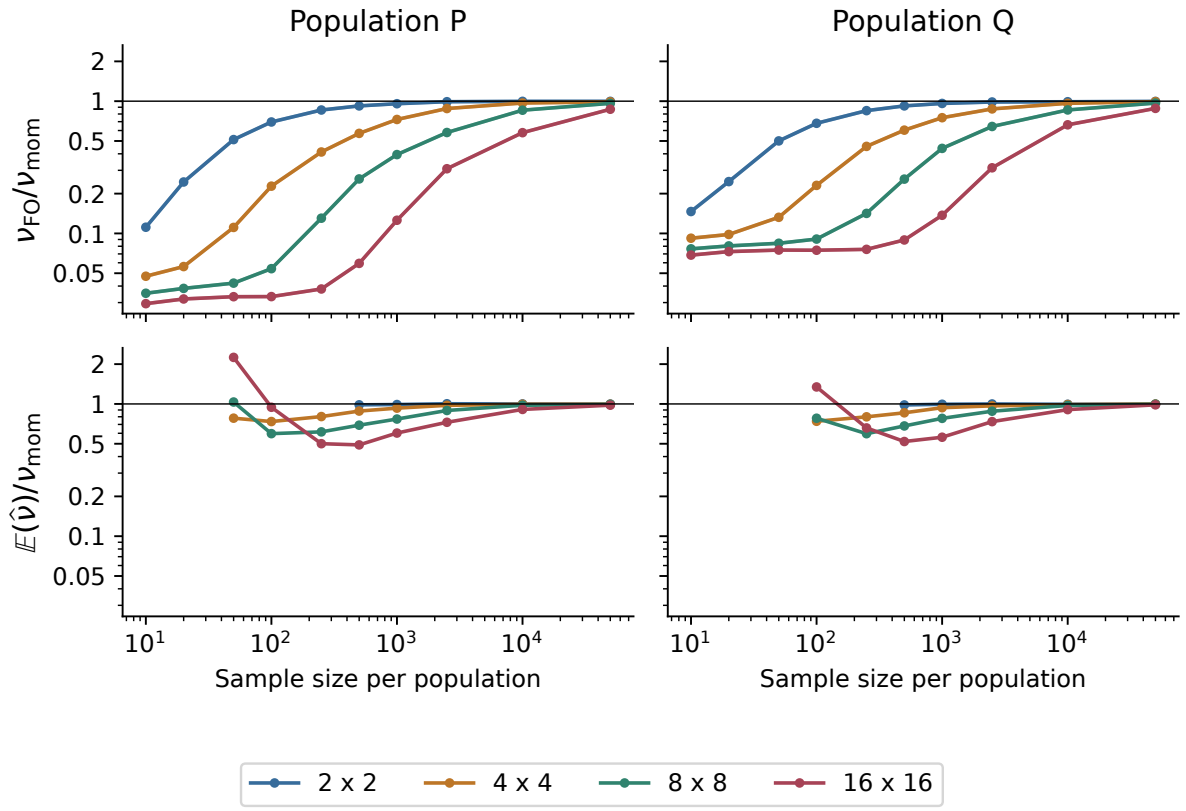


Figure 6.2: Component degrees-of-freedom approximations relative to their moment targets.

In the top row of Figure 6.2, the first-order values remain below the moment targets over most of the sample-size range. The 2×2 curves approach one first, followed by 4×4; the 8×8 and 16×16 curves remain lower for longer. This pattern arises because the two moment errors enter different parts of the degrees-of-freedom calculation. Replacing the enlarged sampling mean of $\widehat{V}(P)$ by the smaller $V(P)$ lowers the squared numerator. A variance prediction below the observed sampling variance lowers the denominator and raises the resulting degrees of freedom; a prediction above the observed variance lowers them. Over most of the range, the reduction from the squared mean outweighs the upward effect of an underestimated denominator. The larger tables have greater mean errors in Figure 6.1, so their first-order degrees of freedom remain farther below the target. As the moment approximations improve, both effects diminish and the ratios approach one. Although the same table-size ordering appears in P and Q, the

larger-table ratios are generally closer to one in Q, whose smaller normalised mean overshoot produces a smaller downward error in the squared numerator.

In the bottom row, the 16×16 curves first appear above the target, fall below it, and then recover towards one. The 4×4 and 8×8 curves have smaller excursions, while the 2×2 curves are already close to one when they become visible. This change from the top row occurs because each sampled table supplies both the squared numerator $\widehat{V}(P)^2$ and the variance prediction $\widehat{\tau}^2(P)/n_P$. At the first shown 16×16 points, the variance prediction is too small on average, tending to raise the estimated degrees of freedom. As the prediction subsequently becomes too large, it tends to lower the estimated degrees of freedom before recovering towards the observed sampling variance. The two inputs come from the same table, so their joint variation also affects the mean degrees of freedom when we average the sampled ratios rather than insert average inputs into a single ratio. With larger samples, both inputs become more stable and the mean estimated degrees of freedom approach the target. They can recover sooner than the population first-order values because their numerators use the sampled $\widehat{V}$, rather than the smaller fixed V. Both populations show these patterns, but for the three larger table sizes the curves first become visible later in Q because the requirement of complete component validity is met at a larger sample size. Missing earlier points indicate failed component calculations in some repetitions, rather than zero degrees of freedom.

A numerical example shows why a fairly accurate variance prediction can still give degrees of freedom below the target. For different-skew 4×4 population P at $n = 1{,}000$ and MI 0.02, the sampling mean of $\widehat{V}(P)$ is about 19% above $V(P)$, whereas $\tau^2(P)/n_P$ is only about 3% below the observed sampling variance. The first-order calculation squares $V(P)$, whereas the moment target squares the larger $\mathrm{E}\{\widehat{V}(P)\}$, so the mean error's downward effect on the numerator is magnified. The slightly underestimated variance denominator offsets only part of this effect, leaving first-order degrees of freedom of 15.30 compared with a moment target of 21.03. Matching these moments chooses the degrees of freedom of the scaled chi-squared approximation; it does not guarantee an exact Student t distribution for the test statistic.

6.1.3 Behaviour near independence

Near independence, the pointwise-MI weights in the linear sampling error become small, so products of cell-probability errors can contribute substantially to sampling variation. These quadratic terms allow sampled tables to show apparent association even when population MI is almost zero. The near-

independence calculation in Appendix A.2 retains them to predict the mean and sampling variance of $\widehat{V}$ and the mean estimated component degrees of freedom.

Figure 6.3 holds the sample size at $n = 1{,}000$ and the table size at 4×4, and increases the common population MI along the horizontal axis. The margins are $m_k(0.7)$ for P and $m_k(0.8)$ for Q. The top row multiplies the observed sampling mean of $\widehat{V}$ by n; the middle row multiplies its observed sampling variance by n^2. Their dashed near-independence predictions use the same scalings and depend on the common table size, sample size and MI, so they are identical in the two columns. The bottom row compares the mean estimated degrees of freedom, its near-independence prediction, the sampling-moment target and the population first-order value. Across the simulation repetitions, we calculate the observed means and variances and average the degrees of freedom estimated separately from each sampled table.

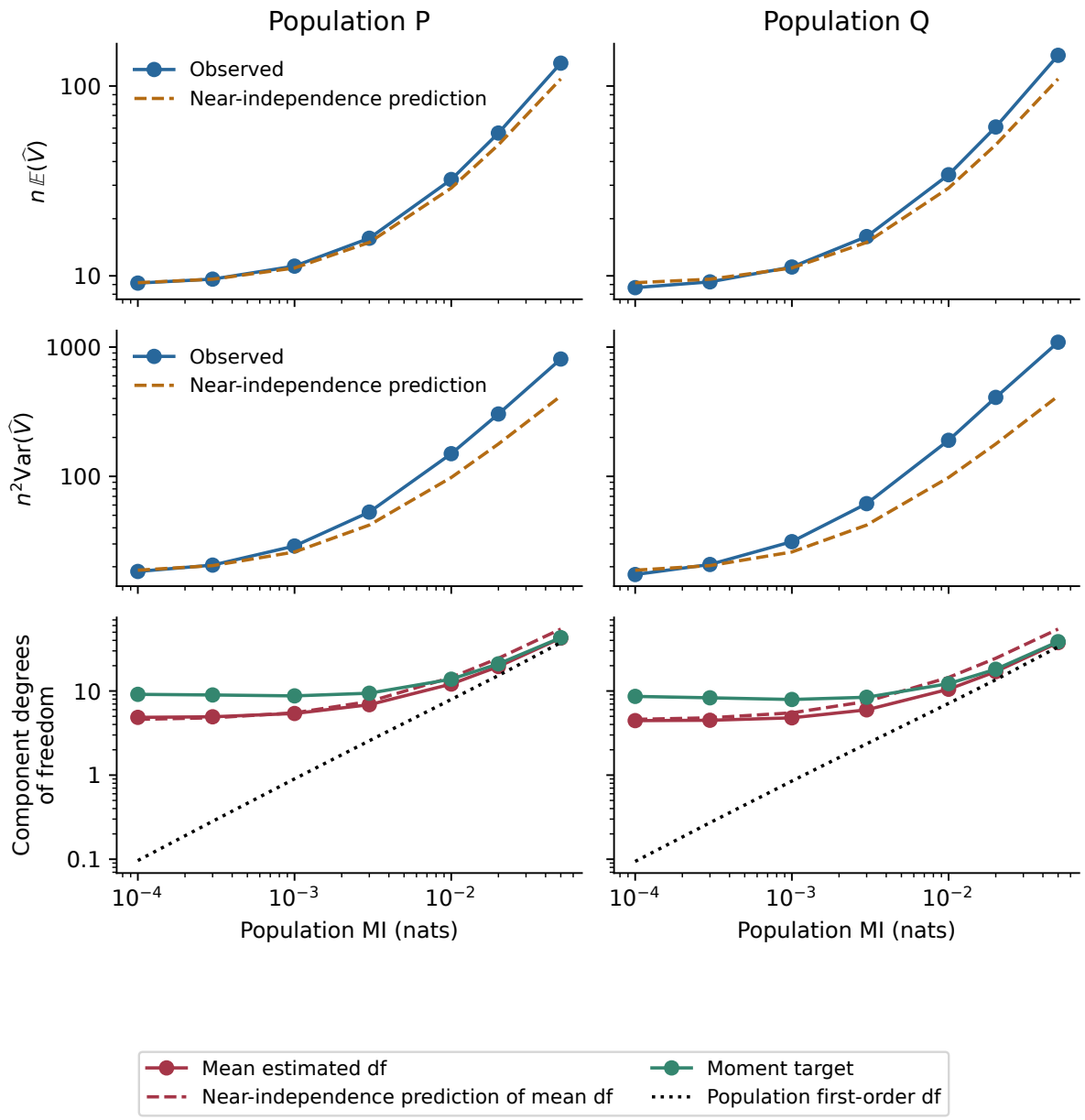


Figure 6.3: Component moments and degrees of freedom near independence.

In the top row of Figure 6.3, the observed scaled mean and its near-independence prediction nearly coincide at the smallest MI values, then both rise as association strengthens, with the observed mean eventually exceeding the prediction. The positive starting level occurs because sampling noise produces positive plug-in MI and $\widehat{V}$ even when population association is almost absent. The local calculation includes this noise contribution through $\widehat{V} \approx 2\widehat{I}$. As MI increases, the cell log ratios move farther from their values at independence, and the omitted terms in the local expansion become more important.

Checks on the sampled tables show that $2\widehat{I}$ then increasingly understates $\widehat{V}$. The observed means also separate between the columns, with Q higher at the larger MI values. The identical local predictions cannot capture this margin dependence because they retain only the common table size, sample size and MI.

In the middle row, the observed scaled sampling variance also starts close to the near-independence prediction, but their separation grows more strongly as MI increases. Checks on the same 4×4, different-skew MI sweep at $n = 1{,}000$ trace this underprediction through the two steps of the local calculation. The calculation first represents $\widehat{V}$ by $2\widehat{I}$, then predicts the sampling variance of that quantity using a noncentral chi-squared distribution. At stronger association, $2\widehat{I}$ varies less between samples than $\widehat{V}$, while the local distribution also understates the observed variation in $2\widehat{I}$ itself. Both errors therefore lower the predicted sampling variance relative to the observed $\mathrm{Var}\{\widehat{V}\}$. The gap is larger than in the mean row in both columns, and the observed sampling variance is greater in Q than in P at the larger MI values. Thus the different margins affect sampling variation beyond the terms retained by the shared local prediction.

In the bottom row, the mean estimated degrees of freedom remain several units at the smallest MI values, around half the sampling-moment target, while the population first-order value approaches zero. This separation arises because the sample-based calculation retains apparent association from sampling noise. Substituting $\widehat{V} \approx 2\widehat{I}$ and $\widehat{\tau}^2 \approx 4\widehat{V}$ into its formula gives $\widehat{\nu}_P \approx n_P \widehat{I}(P)$, whose average stays positive at fixed sample size even when $I(P)$ is almost zero. The population first-order value instead approaches $n_P I(P)$, which becomes very small. The moment target has a shallow initial dip because its observed sampling variance initially grows faster in proportion than its squared sampling mean. Their relative growth later reverses and the target rises, with the mean estimated degrees of freedom approaching it. At stronger association, the dashed prediction rises above the mean estimated degrees of freedom, while the population first-order curve remains below the moment target. Here both local substitutions enter a ratio, with $\widehat{V}$ squared in the numerator and $\widehat{\tau}^2$ in the denominator, so their combined error gives an upward error in the predicted mean, despite the downward moment errors in the upper rows. Both populations show this progression, but Q has lower absolute component degrees of freedom over the displayed range.

The population and sample-based predictions for the sampling variance of $\widehat{V}$ also separate near independence. At the smallest MI target, 0.0001, the population first-order predictions are only about 5% of the observed sampling variances, while the sample-based predictions average roughly twice those variances. The local relation $\widehat{\tau}^2 \approx 4\widehat{V}$ carries sampling noise into the estimated variance prediction, explaining its upward error in this range (Appendix A.2). At MI 0.02, the population first-order variance predictions are

much more accurate, although the sample-based ratios remain above one. The population calculation improves as the linear sampling term becomes more informative, while the local near-independence calculation loses accuracy as association grows. At MI 0.05, the mean and both variance predictions are close to their observed sampling moments, and the mean estimated degrees of freedom are closer to the moment target than the population first-order values. The latter still use V in place of the enlarged sampling mean. Across these MI values, Q has lower absolute component degrees of freedom: its sampling variance is greater relative to its squared sampling mean, giving a lower moment target.

6.2 RQ2: False-positive rates under equal MI

Under $I(P) = I(Q)$, every rejection is a false positive. Both tests use $T = \widehat{\Delta}_{\mathrm{BC}}/\widehat{\mathrm{SE}}$, but Expanded Welch replaces Wald's standard normal cutoff with a larger cutoff from the Student t distribution, reducing rejection at the common nominal p-value threshold. We therefore assess whether the reduction moves the rate towards the nominal 5% level, and use the shared estimates to explain when it helps or increases conservatism.

6.2.1 Overall false-positive patterns

Table 6.1 summarises the comparison across 120 settings at common MI 0.02 nats. A setting is classified as improved when Expanded Welch's false-positive rate is closer to 5% than Wald's by more than 0.3 percentage points, and worsened when it is farther away by more than 0.3 percentage points. Differences within this tolerance are grouped separately. The rows distinguish settings where Wald starts above or below the target, so a reduction in rejection can be assessed against its starting position.

Table 6.1: Change in distance from 5% rejection across the broad null grid.

	Expanded Welch relative to Normal Wald			
Normal Wald rejection rate	Closer to 5%	Farther from 5%	Within tolerance	Total
Above 5%	29	4	5	38
Below 5%	0	48	34	82
Total	29	52	39	120

Where Wald rejects too often, Expanded Welch moves closer to 5% in 29 of 38 settings, although 4 reductions overshoot and leave the rate farther from the target. Where Wald is already conservative, lowering rejection cannot bring the rate closer to 5%, so Expanded Welch either worsens calibration or changes it by no more than the stated tolerance.

Each reported rate divides the number of valid rejections by all simulation repetitions, counting invalid calculations as non-rejections, so the validity fraction is reported separately to distinguish low rejection from a loss of usable calculations. Brackets around rates and shading in rejection plots give pointwise 95% intervals for uncertainty from the finite simulation batch.

Figures 6.4 and 6.5 hold population MI at 0.02 nats and increase the common sample size. Each panel compares Normal Wald with Expanded Welch at one table size, with the black line marking 5%. The first figure uses uniform margins, whereas the second uses $m_k(0.7)$ for P and $m_k(0.8)$ for Q. Vertical scales differ between panels, so comparisons between table sizes require the labelled rates. The lower strips give the fraction of valid outputs for each method.

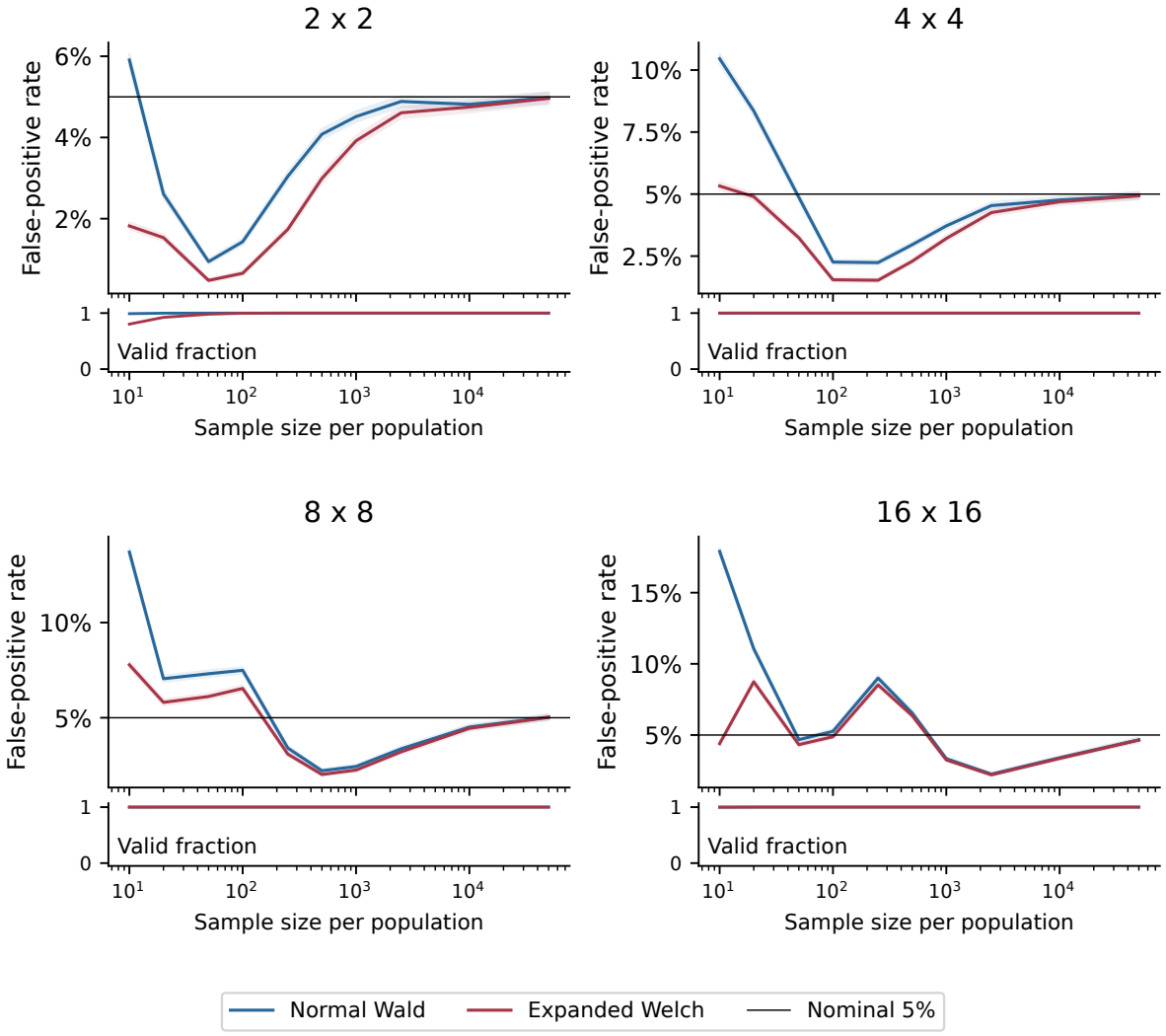


Figure 6.4: False-positive rates under uniform margins.

In the 2×2 panel of Figure 6.4, both rates fall into a deep trough and then recover towards 5%, with Expanded Welch more conservative over most of the range. These uniform-margin settings use identical population tables, $P = Q$, and equal sample sizes, so the expected MI estimation biases cancel in the difference. The trough therefore requires an explanation through the standard error and the distribution of T. Because sampled tables with larger absolute MI differences also tend to give larger standard errors, division by those denominators reduces the extreme values of T. This reduction in extreme statistics contributes to the low rejection rate, which Expanded Welch's larger cutoff lowers further. The dependence weakens at larger sample sizes, allowing both rates to recover, as shown by the diagnostics in Section 6.2.2.

In the 4×4 panel, Wald initially rejects too often while Expanded Welch is close to the target, before both rates fall below 5% and then recover. This reversal accompanies the average $\widehat{\mathrm{SE}}^2$ moving from below the observed variance of $\widehat{\Delta}_{\mathrm{BC}}$ at small sample sizes to above it over the conservative range. For a given difference estimate, a smaller standard error increases $|T|$, while a larger standard error decreases it. The same increase in rejection cutoff therefore helps in the first range and deepens conservatism in the second. With larger samples, the squared standard error becomes more accurate and the methods approach 5%.

In the 8×8 panel, an early shoulder of excessive rejection is followed by a later conservative trough and recovery. The shared squared standard error initially underestimates the difference's sampling variance, then overestimates it across the trough. The larger alphabet spreads probability over more combinations, so the sampled tables require more observations before these variance errors diminish. Expanded Welch reduces the early excess but also lowers the conservative part of the curve. As in 4×4, its effect on calibration changes along the horizontal axis.

The 16×16 panel has an additional rise between the early decline and the later conservative trough. The average variance error also changes direction more than once as the sampled tables fill: underestimation, closer agreement, renewed underestimation and then overestimation accompany the corresponding changes in rejection. Because the populations are identical, expected difference bias still cancels, so the nonmonotonic curve reflects errors in the standard error and the tails of T, rather than a changing population difference. Expanded Welch lowers rejection throughout, but the methods become close well before both rates settle near 5%, so their agreement alone does not establish accurate false-positive rates.

Under different-skew margins, $I(P) = I(Q)$ in Figure 6.5, but the population tables differ and their expected MI estimation biases can therefore differ too. The resulting bias in $\widehat{\Delta}_{\mathrm{BC}}$ can shift the shared statistic away from zero as well as changing its spread.

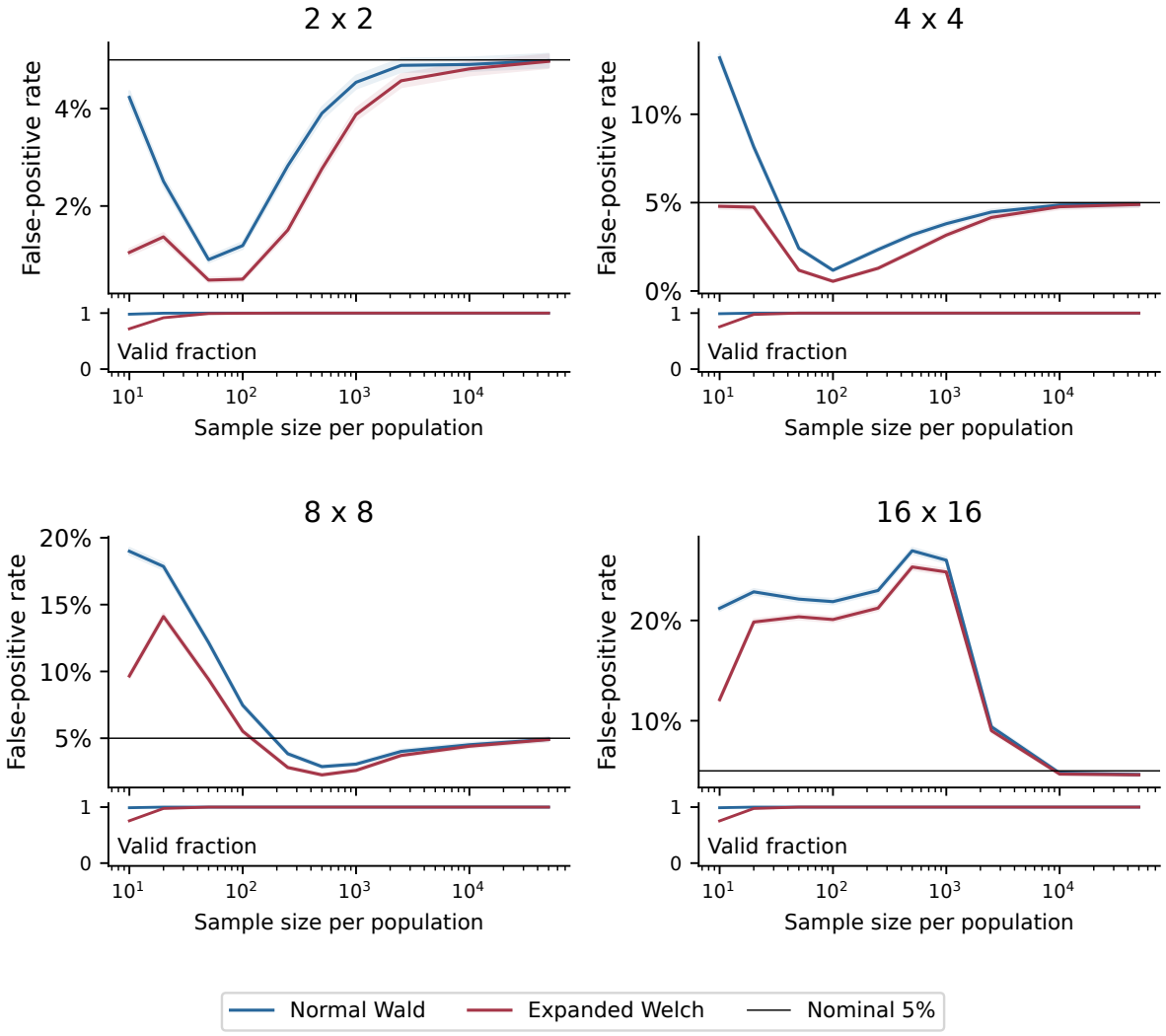


Figure 6.5: False-positive rates under different-skew margins.

In the different-skew 2×2 panel, Wald remains below 5% throughout the grid, so Expanded Welch's reduction mainly increases conservatism. Both methods again have a trough and recovery, retaining the small-table pattern seen under uniform margins despite the changed margins. As the shared estimates improve, both rates recover towards 5%, while the increasing degrees of freedom bring Expanded Welch's cutoff closer to Wald's.

In 4×4, Wald starts above the target and then falls below it, while Expanded Welch lowers both regions. The average squared standard error moves from underestimating to overestimating the observed difference variance, helping explain the fall into the trough. As that overestimation diminishes, both rates rise towards 5%. At $n = 10$, the first Expanded Welch rate is close to 5% even though only about three

quarters of calculations are valid. Its apparent closeness at that point therefore includes repetitions recorded as non-rejections because the calculation failed.

The 8×8 panel begins with much greater over-rejection than its uniform counterpart. Expanded Welch initially rises as its validity and cutoff change, then both methods fall below 5% before recovering. The diagnostics show that positive bias moves the centre of $\widehat{\Delta}_{\mathrm{BC}}$ away from zero, while overestimation of $\widehat{\mathrm{SE}}^2$ in the intermediate range tends to reduce $|T|$. At $n = 100$, both effects are present even though Expanded Welch brings the rate closer to 5%. As the bias diminishes, the remaining variance overestimation accompanies the conservative trough. The larger cutoff deepens that trough before the shared estimates recover. A rate close to the target at an intermediate point can therefore reflect the combined effects of errors rather than accurate estimation of each input.

In the 16×16 panel, both methods severely over-reject across several intermediate sample sizes. Here the bias in $\widehat{\Delta}_{\mathrm{BC}}$ grows relative to its sampling SD, shifting the statistic away from zero (see Section 6.2.2 for the comparison with the unbiased same-skew case). The squared standard error is also too large on average over part of this range, but that compensating effect does not remove the large displacement. Expanded Welch changes the cutoff without changing the displaced difference estimate, so its reduction in rejection leaves substantial over-rejection. With larger samples, bias diminishes and both rates fall towards 5%.

The validity strips separate early calculation failures from these later patterns. Under uniform margins, appreciable losses are concentrated in the smallest 2×2 samples, whereas under different skew several table sizes lose outputs at the smallest sample sizes. Across the full 120-setting grid, the minimum valid fractions are 0.958 for Normal Wald and 0.631 for Expanded Welch. The conservative troughs in 4×4 and larger tables, and the different-skew 16×16 plateau, persist after validity is complete, so those patterns require explanations through the estimates and statistic.

At the largest sample size in both figures, both tests' rates are close to 5% across the table sizes. More accurate cell frequencies reduce the bias and variance errors entering the shared statistic, while larger estimated degrees of freedom bring Expanded Welch's cutoff towards Wald's. The curves consequently approach one another and the nominal target in these settings.

6.2.2 Bias and standard-error accuracy

RQ1 examined the moments of $\widehat{V}(P)$ and $\widehat{V}(Q)$ needed to choose degrees of freedom. For the shared statistic, we now separately check whether $\widehat{\Delta}_{\mathrm{BC}}$ is centred on the population difference and whether $\widehat{\mathrm{SE}}^2$ describes its sampling variance. At small sample sizes, $V(P)/n_P$ can itself understate MI sampling variance because it includes only linear sampling error. Consequently, $\widehat{V}(P)/n_P$ can exceed that population approximation while still falling below the actual sampling variance of the MI estimate.

Table 6.2 calculates both diagnostics across the evaluation repetitions. "Bias / SD" divides $\mathrm{E}(\widehat{\Delta}_{\mathrm{BC}}) - \Delta$ by the observed sampling SD of $\widehat{\Delta}_{\mathrm{BC}}$, so it measures displacement relative to the spread of the estimates. The variance ratio divides the average $\widehat{\mathrm{SE}}^2$ by the observed $\mathrm{Var}(\widehat{\Delta}_{\mathrm{BC}})$: values below one indicate underestimation of sampling variance, and values above one indicate overestimation. The observed variance is measured about the sampling mean, so this ratio can be close to one even when the difference is biased. The table compares the selected null settings A, uniform 4×4 with $n = 20$; B, different-skew 8×8 with $n = 100$; and C, different-skew 4×4 with $n = 10{,}000$, together with the sparse 16×16 cases at $n = 1{,}000$. All use equal sample sizes and common MI 0.02 nats.

Table 6.2: Bias and standard-error accuracy at selected equal-MI settings.

Setting	Bias / SD	$\mathrm{E}(\widehat{\mathrm{SE}}^2)/\,\mathrm{Var}(\widehat{\Delta}_{\mathrm{BC}})$
A: uniform 4×4, $n = 20$	0.003	0.827
B: different skew 8×8, $n = 100$	0.870	1.535
C: different skew 4×4, $n = 10{,}000$	0.000	1.010
16×16, same skew, $n = 1{,}000$	-0.003	1.929
16×16, different skew, $n = 1{,}000$	2.058	1.938

Table 6.3 gives rejection rates for the same five settings, followed below the divider by the same 16×16 populations at $n = 10{,}000$. Each rate divides valid rejections by all 100,000 repetitions and is reported with a pointwise 95% interval. Both tests are valid in every repetition at these seven settings.

Table 6.3: Null rejection rates with pointwise 95% intervals.

Setting	Normal Wald	Expanded Welch
A	8.35% [8.18, 8.53]	4.90% [4.76, 5.03]
B	7.47% [7.30, 7.63]	5.52% [5.38, 5.67]
C	4.87% [4.73, 5.00]	4.77% [4.64, 4.90]
16×16, same skew, $n = 1{,}000$	0.60% [0.56, 0.65]	0.53% [0.48, 0.57]
16×16, different skew, $n = 1{,}000$	26.01% [25.74, 26.28]	24.84% [24.57, 25.10]
16×16, same skew, $n = 10{,}000$	4.38% [4.26, 4.51]	4.26% [4.14, 4.39]
16×16, different skew, $n = 10{,}000$	4.78% [4.65, 4.91]	4.68% [4.55, 4.82]

At A, difference bias is negligible, but the average $\widehat{\mathrm{SE}}^2$ is below the observed $\mathrm{Var}(\widehat{\Delta}_{\mathrm{BC}})$, tending to enlarge $|T|$ and accompanying Wald's excessive rejection. Expanded Welch's larger cutoff brings the rate close to 5%. At B, $\widehat{\Delta}_{\mathrm{BC}}$ is biased upwards while $\widehat{\mathrm{SE}}^2$ overestimates its sampling variance. The upward displacement tends to increase rejection, whereas the larger denominator tends to reduce it. Expanded Welch lowers rejection further but remains above 5%, so its improvement occurs with different input errors from those at A. At C, bias is negligible and the variance ratio is close to one, accompanying rates close to 5% for both methods.

The same margins and sample sizes need not give the same calibration at stronger association. Table 6.4 retains A's and B's margins, table sizes and sample sizes while increasing their common population MI. Rates and their pointwise 95% intervals are percentages of all 100,000 repetitions.

Table 6.4: False-positive rates at stronger common MI.

Setting	Common MI (nats)	Normal Wald	Expanded Welch
A	0.3299	16.16 [15.94, 16.39]	13.56 [13.35, 13.78]
B	0.2806	7.62 [7.46, 7.79]	7.25 [7.09, 7.41]

At A, increasing the common MI raises both rejection rates well above their values at MI 0.02, and Expanded Welch no longer brings rejection close to 5%. At B, Wald's rate changes little, but Expanded Welch's earlier reduction becomes much smaller. Both methods over-reject at both stronger-MI settings

even though the larger cutoff still lowers rejection. The improvement at MI 0.02 is therefore specific to that tested population pair rather than a property of its table dimensions and sample size alone.

The sparse different-skew case shows why changing the cutoff cannot resolve a displaced difference estimate. At 16×16 and $n = 1{,}000$, the difference bias is about 0.030 nats, or 2.06 observed SDs. Under the equal-MI null, its sampling centre $\mathrm{E}(\widehat{\Delta}_{\mathrm{BC}}) = \Delta + \mathrm{Bias}(\widehat{\Delta}_{\mathrm{BC}})$ becomes

$$\mathrm{E}(\widehat{\Delta}_{\mathrm{BC}}) \approx 0 + 0.02985 = 0.02985 \text{ nats.} \tag{6.1}$$

With an observed SD of approximately 0.01450 nats, this positive shift places the sampling distribution largely above zero and moves most rejections into the positive tail despite a zero population difference. Expanded Welch leaves this centre unchanged, so both tests reject far too often.

The same-skew 16×16 case has almost the same variance ratio but rejection far below 5%. Under equal MI, the expected estimation biases cancel in the difference because the population tables and sample sizes are identical. Its average $\widehat{\mathrm{SE}}^2$ exceeds the observed $\mathrm{Var}(\widehat{\Delta}_{\mathrm{BC}})$, shrinking $|T|$ without a large bias shifting the estimates away from zero. Under different skew, that variance overestimation coexists with the large positive bias, and rejection instead remains excessive. Thus the similar variance ratios in Table 6.2 can accompany rates on opposite sides of 5% in Table 6.3. At $n = 10{,}000$, both skew regimes have rejection rates closer to 5%.

For the identical uniform-margin populations, expected difference bias cancels, while changes in the average $\widehat{\mathrm{SE}}^2$ below or above the observed $\mathrm{Var}(\widehat{\Delta}_{\mathrm{BC}})$ tend to enlarge or reduce $|T|$, respectively. These changes accompany the repeated crossings in the 16×16 panel of Figure 6.4. The estimates recover at different rates as sparse tables fill, so increasing sample size need not improve rejection monotonically.

The 2×2 trough also involves how the numerator and denominator move together within repetitions. Larger absolute MI differences tend to occur with larger standard errors, so division suppresses the extreme values that would otherwise cause rejection. Consequently, rejection can be far below 5% even when the average $\widehat{\mathrm{SE}}^2$ is fairly close to the observed difference variance. Beyond the trough, this dependence weakens and the tails of T move closer to the normal approximation. The Spearman rank correlation between $|\widehat{\Delta}_{\mathrm{BC}} - \mathrm{mean}(\widehat{\Delta}_{\mathrm{BC}})|$ and $\widehat{\mathrm{SE}}$, calculated over repetitions with both quantities finite, falls from 0.656 at $n = 50$ to 0.167 at $n = 1{,}000$. Together these checks explain the broad patterns without isolating every finite-sample contribution to rejection.

6.3 RQ3: Power at a fixed MI difference

The power study retains the four table sizes, three margin regimes and ten sample sizes of the broad null comparison, but fixes $I(P) = 0.02$ and $I(Q) = 0.03$ nats. The true difference is therefore $\Delta = -0.01$ nats throughout. Each plotted rate divides valid rejections by all 100,000 repetitions at that setting, with invalid calculations counted as non-rejections. Shading gives pointwise 95% intervals, and the lower strips show the fraction of valid results.

Both tests use their usual p-value threshold of 0.05. Expanded Welch's larger cutoff makes its rejections a subset of Wald's, so the comparison quantifies how much detection probability is lost. These rates are read alongside the false-positive results in Section 6.2.1: high rejection under an alternative can coexist with excessive rejection under equal MI. Because the tests are two-sided, power also includes rejections whose estimated MI difference has the opposite sign to the population difference.

6.3.1 Uniform margins

Figure 6.6 uses uniform row and column margins in both populations. Each panel holds the population tables and MI difference fixed while increasing the common sample size along the logarithmic horizontal axis. The two lines compare Normal Wald and Expanded Welch at the same sampled table pairs.

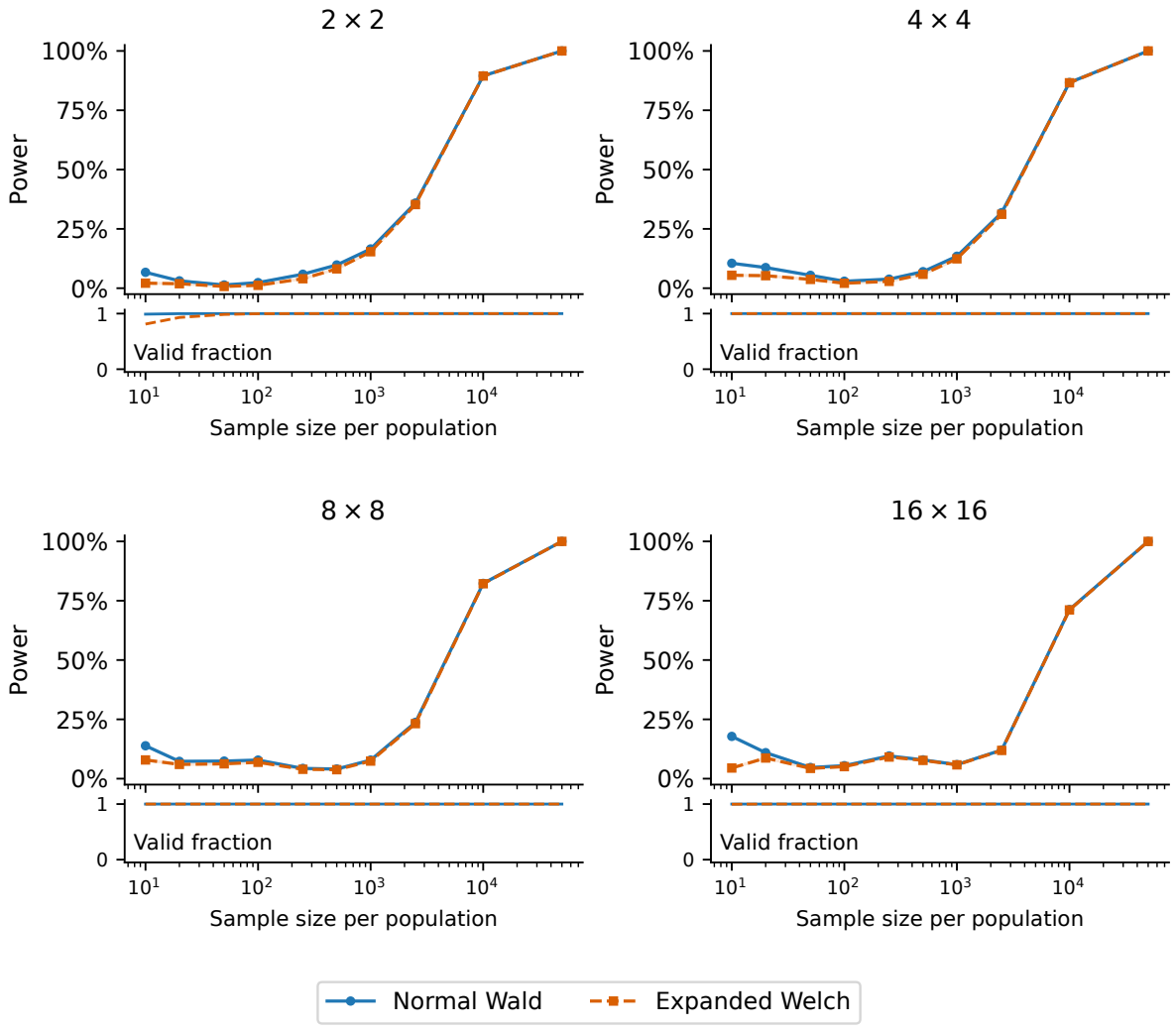


Figure 6.6: Power at a fixed MI difference under uniform margins.

All four panels show low detection over part of the small-sample range, followed by a sustained rise with more observations. Larger tables require more observations before power becomes high. For example, at $n = 10{,}000$, Expanded Welch detects the difference in 89.44% of 2×2 repetitions and 71.12% of 16×16 repetitions. This ordering reflects both the sparse-table estimation errors discussed in Section 6.2.2 and the population variances: in this copying family, increasing table size also increases $V(P)$ and $V(Q)$. Their contributions $V(P)/n + V(Q)/n$ therefore give a larger first-order sampling variance for the same MI gap and sample size. As the variance decreases with more observations, the estimated difference becomes more consistently separated from zero and power rises.

Expanded Welch remains below Wald, with the largest separation at the smallest sample size in the 16×16 panel. Its valid fraction there is above 99%, so most of that separation comes from the larger cutoff rather than failed calculations. The method gap becomes small in the larger-sample range as increasing degrees of freedom bring the Student t distribution close to the standard normal distribution. The early shoulders and troughs also occur in the null rejection curves, where standard-error errors and the distribution of T can produce excessive or conservative rejection. Consequently, the early high points are not evidence that fewer observations reliably identify the MI difference.

6.3.2 Same-skew margins

Figure 6.7 assigns both populations the row and column profile $m_k(0.8)$, retaining the same MI targets and sample sizes. The margins agree between populations, but their different MI values require different copying probabilities and hence different population tables.

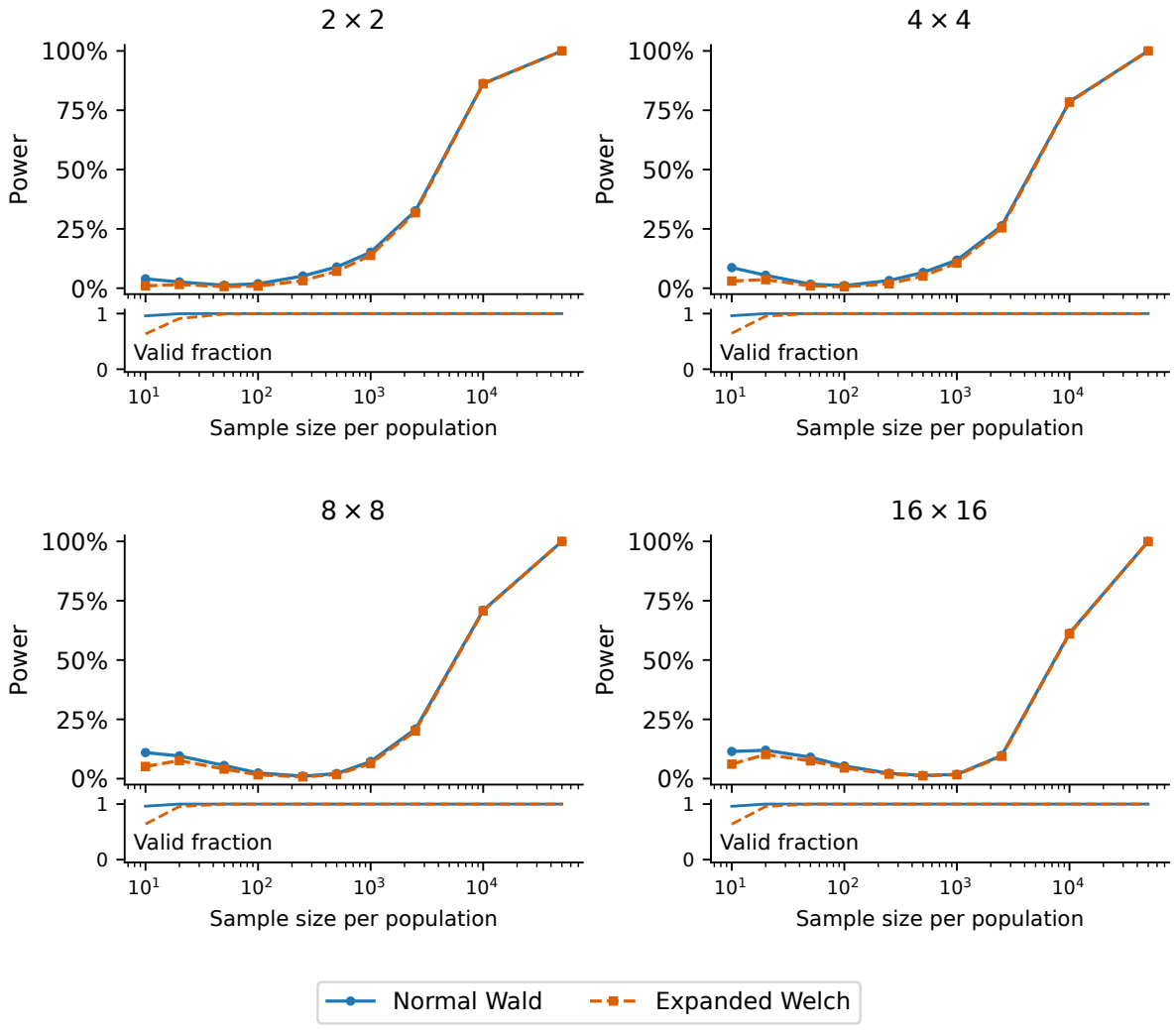


Figure 6.7: Power at a fixed MI difference under same-skew margins.

The curves again have an early low-power range followed by recovery, with the trough occurring later for larger tables. Relative to uniform margins, the uneven frequencies leave more combinations rarely observed, and the population pointwise-MI variances are also larger. Both features make the fixed gap harder to detect. In the 16×16 case at $n = 1{,}000$, the average squared standard error is 1.81 times the observed variance of the difference, while difference bias is small relative to the true gap. The oversized denominator suppresses $|T|$, contributing to Expanded Welch's power of only 1.64% at that setting.

Both methods gain power once larger samples improve the cell estimates and reduce the spread of the MI difference. Expanded Welch reaches 60.99% in 16×16 at $n = 10{,}000$, compared with 71.12% under uniform margins, and exceeds 99.9% in every same-skew panel at $n = 50{,}000$. The differences between

methods become much smaller than these differences between table sizes and margin regimes. At the smallest sample sizes, however, the validity strips show that failed Expanded Welch calculations also contribute to its lower all-repetition power; the larger cutoff is not the only source of that loss.

6.3.3 Different-skew margins

Figure 6.8 uses $m_k(0.7)$ for P and $m_k(0.8)$ for Q. The true difference remains -0.01 nats, but the unequal margins can give the MI estimates different residual biases, shifting the centre of their difference.

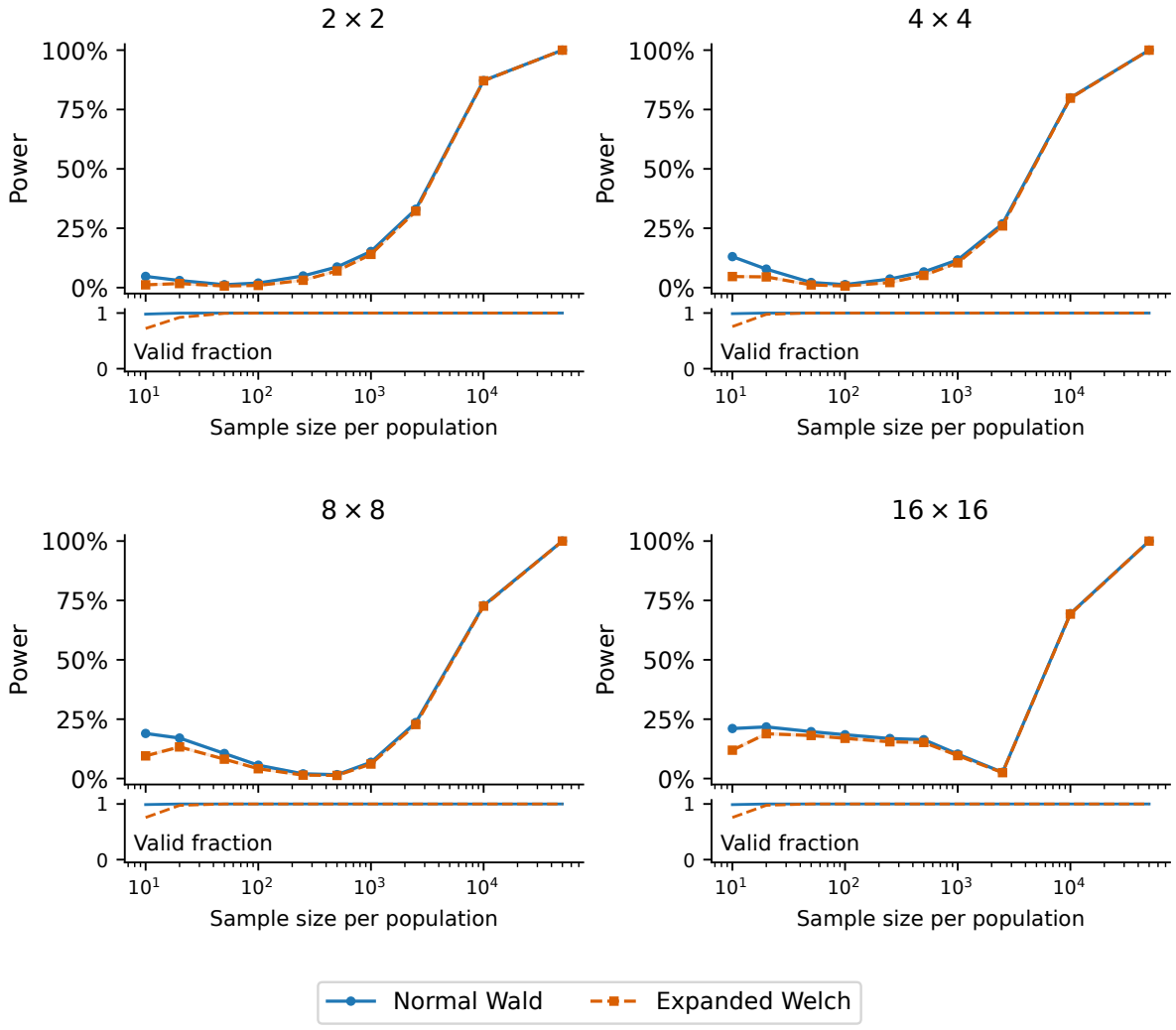


Figure 6.8: Power at a fixed MI difference under different-skew margins.

The 2×2 and 4×4 panels retain the low-power range and subsequent rise, while the two larger tables have more pronounced early rejection before falling and recovering. The 16×16 curve shows why this early rejection needs careful interpretation. At $n = 1{,}000$, positive difference bias is approximately 0.0307 nats, moving the sampling mean from the true -0.01 to approximately $+0.0207$ nats. Expanded Welch rejects in 9.66% of repetitions, but more than 99.9% of those rejections have a positive estimated difference, opposite to the true direction. The shared biased numerator therefore produces rejection without reliably identifying which population has greater MI, as in the displaced null comparison in Section 6.2.2.

As sample size increases, that positive bias decreases and the sampling centre passes back towards zero. At $n = 2{,}500$, the remaining positive bias nearly cancels the negative population difference, leaving the sampling mean close to zero, and Expanded Welch's power falls to 2.48%. With still more observations, the bias diminishes further and the sampling centre approaches the negative population difference, while sampling variance decreases. Negative-tail rejection then rises and power becomes high. Changing the cutoff lowers both parts of this nonmonotonic curve without removing the biased movement of its centre.

Across all three margin regimes, both methods exceed 99.9% power at $n = 50{,}000$. At $n = 10{,}000$, their power differences are below 0.32 percentage points at every table size and margin regime. The fixed-gap study therefore shows a detection cost from Expanded Welch's larger cutoff, especially with small samples, alongside shared estimation errors that can leave both methods with low power or rejection in the wrong direction. The false-positive findings determine whether that lower rejection improves calibration in the corresponding null setting.

6.4 Application to online shopping data

The Online Shoppers Purchasing Intention dataset records website sessions and whether each session ends in a purchase (Sakar and Kastro, 2018). We compare month–purchase MI between new and returning visitors, then make the same comparison for weekend–purchase MI. For each predictor, X is month or weekend status and Y is the purchase outcome, giving joint distributions P and Q of (X, Y) among new and returning visitors, respectively. Each comparison therefore tests $H_0 : I(P) = I(Q)$ for that predictor and purchase outcome.

The dataset contains 12,330 sessions, each belonging to a different user according to the source documentation (Sakar and Kastro, 2018). We retain all 1,694 sessions labelled `New_Visitor` and 10,551 labelled `Returning_Visitor`, excluding the 85 `Other` sessions. The outcome is the original Boolean `Revenue` variable, which is true when a purchase occurs and false otherwise. We treat the two visitor groups as independent samples and the sessions within each group as independent observations.

January and April have no recorded sessions, so each month table has 10×2 cells, with rows for the ten recorded months and columns for no purchase or purchase. All ten months occur in both visitor groups, although the new-visitor February row contains only one session and no purchase. Both groups use the same categories, retaining zero cells without smoothing. Replacing month with the Boolean `Weekend` variable gives a 2×2 table for each group, with weekday and weekend rows and all four cells observed.

These two shopping comparisons were selected from six comparisons across three public datasets. The other four concern occupation and education in Adult (Becker and Kohavi, 1996), and study time and maternal education in Student Performance (Cortez, 2008). To account for this exploratory family, we apply Holm's adjustment separately to each method's six p-values (Holm, 1979). Table B.1 in Appendix B.7 lists the complete family and the unadjusted p-values used in the calculation.

The corrected difference $\widehat{\Delta}_{\mathrm{BC}}$ is new-visitor MI minus returning-visitor MI, so a negative value indicates higher estimated MI among returning visitors. The leading MI bias correction subtracts $(10-1)(2-1)/(2n) = 9/(2n)$ from each month plug-in estimate (Miller, 1955; Moddemeijer, 1989), and $1/(2n)$ from each weekend estimate, where n is the relevant group size. Table 6.5 gives the corrected MI estimates, their differences and shared standard errors, $\widehat{\mathrm{SE}}$, all in nats, alongside the resulting $T = \widehat{\Delta}_{\mathrm{BC}}/\widehat{\mathrm{SE}}$.

Table 6.5: Corrected MI estimates and uncertainty of their differences.

	Corrected MI				
Predictor	New	Returning	$\widehat{\Delta}_{\mathrm{BC}}$	$\widehat{\mathrm{SE}}$	T
Month	0.004316	0.017880	-0.013563	0.003333	-4.069
Weekend	0.000664	0.000727	-0.000063	0.001124	-0.056

For month, returning visitors have the higher corrected MI estimate, giving a negative difference whose magnitude is about 4.07 times its estimated standard error:

$$T \approx \frac{-0.013563}{0.003333} \approx -4.069. \tag{6.2}$$

Both tests use this statistic, but Normal Wald compares it with the standard normal distribution, whereas Expanded Welch uses a Student t distribution with estimated degrees of freedom. Table 6.6 gives Expanded Welch's degrees of freedom and the unadjusted two-sided p-values, which we compare with the usual p-value threshold of 0.05.

Table 6.6: Two-sample test results for online shopping sessions.

		p-value	
Predictor	Expanded Welch degrees of freedom	Normal Wald	Expanded Welch
Month	26.79	4.71×10^{-5}	3.73×10^{-4}
Weekend	2.18	0.9554	0.9601

Both methods reject equal month–purchase MI because their p-values are below 0.05. Expanded Welch gives a larger p-value because the Student t distribution with 26.79 degrees of freedom assigns more probability to statistics at least this far from zero than the standard normal distribution. The Holm-adjusted month p-values remain below 0.05 at 0.000189 for Normal Wald and 0.00149 for Expanded Welch, so the evidence of a difference persists after accounting for the six comparisons.

For weekend, both corrected MI estimates are approximately 0.0007 nats, and their difference is only about 0.056 times its estimated standard error. The statistic therefore lies close to zero under either reference distribution, so both p-values are near one and neither test rejects equal weekend–purchase MI. This non-rejection does not establish equality. Expanded Welch's 2.18 degrees of freedom remain small despite the large samples because $\widehat{V}(P)$ and $\widehat{V}(Q)$ have substantial estimated fluctuations relative to their magnitudes. Equation (4.44) compares each squared estimate with its estimated sampling variance, so greater relative uncertainty gives fewer component degrees of freedom, as illustrated near independence in Figure 6.3.

Because the population tables are unknown, we cannot directly check the estimated degrees of freedom against the sampling-moment target or measure false-positive rates.

Chapter 7

Conclusions

7.1 Contributions and main findings

This thesis derives an Expanded Welch–Satterthwaite test for comparing categorical MI across independent populations by calculating the observation-level sensitivity of the estimated pointwise-MI variance. This sensitivity accounts for changes in cell probabilities, margins and pointwise MI. The variance of this sensitivity gives the first-order approximation $\mathrm{Var}\{\widehat{V}(P)\} \approx \tau^2(P)/n_P$, which moment matching uses to choose MI-specific degrees of freedom for a Student t distribution.

7.1.1 RQ1: Accuracy of variance approximations

The predicted sampling variance of $\widehat{V}(P)$ can be close to its observed value even when the sampling mean $\mathrm{E}\{\widehat{V}(P)\}$ remains above $V(P)$. Because the moment target in Equation (5.5) squares the mean in its numerator, replacing that sampling mean by the smaller population value lowers $\nu_{P,\mathrm{FO}}$ relative to $\nu_{P,\mathrm{mom}}$, even when the sampling-variance approximation is accurate. The sampling mean $\mathrm{E}(\widehat{\nu}_P)$ can also fall below the moment target, with the size of both discrepancies depending on the population and sample size.

Near independence, the first-order prediction $\tau^2(P)/n_P$ for $\mathrm{Var}\{\widehat{V}(P)\}$ is small and quadratic sampling terms remain substantial. The calculation in Appendix A.2 explains the observed scale of the estimated component degrees of freedom. These results explain why the first-order approximation becomes

inadequate near independence and why substituting sample probabilities requires its own evaluation, even when the population approximation is accurate.

7.1.2 RQ2: False-positive rates under equal MI

At common MI 0.02 nats, Expanded Welch brings false-positive rates closer to 5% at A (uniform 4×4, $n_P = n_Q = 20$) and B (different-skew 8×8, $n_P = n_Q = 100$), as defined in Table 6.2. Both tests return valid results in every repetition at these settings, so the lower rejection rates reflect the larger cutoff from the Student t distribution. Across the broad grid, Expanded Welch also lowers rates that are already below 5%, increasing conservatism. The larger cutoff therefore helps when it corrects excessive rejection but worsens calibration when rejection is already too infrequent.

A false-positive rate near 5% does not establish that the estimated component degrees of freedom match their sampling-moment targets. At A, the average $\widehat{\mathrm{SE}}^2$ is below the observed $\mathrm{Var}(\widehat{\Delta}_{\mathrm{BC}})$, while the low estimated degrees of freedom increase the rejection cutoff, so the two errors can compensate in the false-positive rate. At B, the average squared standard error exceeds the observed variance of the MI difference and the difference is biased, so the same explanation does not apply unchanged.

In the sparse same-skew null, identical population tables at equal sample sizes have the same expected MI estimation bias, so that bias cancels in the difference. The average $\widehat{\mathrm{SE}}^2$ is nevertheless almost twice the observed $\mathrm{Var}(\widehat{\Delta}_{\mathrm{BC}})$ in the sparse copying case, and both tests are conservative. Unequal sample sizes can also leave unequal residual biases in otherwise identical populations.

Under different-skew margins, the large positive bias in the estimated MI difference instead accompanies excessive rejection by both tests in the sparse 16×16 null comparison (Section 6.2.2).

7.1.3 RQ3: Power at a fixed MI difference

The power study uses the full 120-setting grid with $I(P) = 0.02$ and $I(Q) = 0.03$ nats. At the usual p-value threshold of 0.05, Expanded Welch's rejections remain a subset of Wald's, so its lower rejection probability also reduces power. The size of that loss depends on the population and sample size, with failed calculations contributing at the smallest samples. At $n = 10{,}000$, the power gap is below 0.32 percentage points throughout the grid, and both methods exceed 99.9% power at $n = 50{,}000$.

High rejection in a sparse alternative does not necessarily identify the direction of the MI difference reliably. In the different-skew 16×16 case at $n = 1{,}000$, positive estimation bias outweighs the true negative difference, and almost all rejections have the opposite estimated direction. As that bias diminishes, the sampling centre first approaches zero, reducing rejection, before moving towards the negative population difference and giving higher power. Expanded Welch's cutoff changes the frequency of rejection but leaves this movement of the shared numerator unchanged.

The observation-level sensitivity supplies an MI-specific basis for choosing the cutoff from the Student t distribution, while the simulations distinguish errors in the degrees-of-freedom approximations from errors in the shared statistic. Expanded Welch brings false-positive rates closer to 5% in selected tested settings and reduces detection probability at the same nominal threshold. Where Wald is already conservative, the larger cutoff instead lowers both null rejection and power. These findings support improvements in particular settings while showing the detection cost and estimation errors that limit the method.

7.2 Limitations

The derivation assumes independent categorical observations, fixed categories, positive population cell probabilities, and positive $V(P)$, $V(Q)$, $\tau^2(P)$ and $\tau^2(Q)$. Paired samples would require a covariance term in the difference variance. Growing alphabets and continuous MI estimators have different approximation errors. Exact independence requires the second-order argument in Appendix A.2.

The findings concern the tested copying populations. The power study holds $I(P) = 0.02$ and $I(Q) = 0.03$ nats throughout the grid, so it evaluates one absolute MI gap with Q having greater MI. The copying construction uses identical row and column margin profiles within each table and places its added association on matching diagonal cells in both populations. Other association constructions, directions and magnitudes of MI change, unequal row and column margins within a table and other locations of dependence remain untested by the power study.

All reported simulations use equal sample sizes, for which the leading bias corrections cancel from the MI difference. They therefore do not evaluate the correction's performance under unequal sample sizes, and residual biases can still differ between populations in the tested equal-sample settings.

The power comparison uses the proposed tests' nominal thresholds, so their false-positive rates need not match. A larger rejection probability under the alternative therefore cannot establish superior power at a

common achieved false-positive rate. The pointwise confidence intervals quantify simulation uncertainty for each specified population and sample size.

The online-shopping application illustrates how to report the estimated MI difference and its direction, standard error, p-value and Expanded Welch degrees of freedom. Its unknown population MI values prevent a direct check of false-positive rates. The application therefore demonstrates interpretation of the test outputs, while the simulation evidence establishes their behaviour under known populations.

7.3 Future work

A first extension is to derive the second-order mean contribution to the variance estimator and evaluate a corresponding degrees-of-freedom correction. Figure 6.3 shows that the mean approximation's error changes with MI and differs between the populations, so the correction should account for the population table. Any estimated correction should then be evaluated through its effect on the test's false-positive rates and power.

Improved MI bias and standard-error estimators could address the sparse-table failures. Their effects should be assessed separately by changing one estimate while holding the others fixed, then measuring null rejection, power and estimated direction. Near independence, the quadratic sampling approximation may support a more accurate comparison of MI differences. Any likelihood-ratio or resampling procedure must preserve the equal-MI null even when the complete population tables differ.

Reversed power paths, stronger dependence at small sample sizes, alternative association constructions, unequal margins and unequal sample sizes would measure how much the false-positive rates and power change across a wider range of populations. Simulations from plausible application populations could also examine sensitivity to rare cells and distinct equal-MI tables, with conclusions tied to the populations chosen for that assessment.

Appendix A

Near-Independence Calculations

These calculations support the near-independence results in Chapter 4 and their interpretation in Chapter 6. The first calculation varies population association while keeping the margins fixed. The second retains the quadratic sampling variation needed at independence and very small MI.

A.1 Fixed-margin behaviour near independence

Let $P_0 = (q_{ij})$ be independent, with positive cells $q_{ij} = a_i b_j$. Consider $P_t = P_0 + tH$, where every row and column sum of H is zero. For sufficiently small t, all cells remain positive and the margins remain a and b. We expand around $t = 0$, where $I(P_0) = 0$.

For $f(t) = I(P_t)$, the second-order Taylor formula is

$$f(t) = f(0) + f'(0)t + \frac{f''(0)}{2!}t^2 + O(t^3). \tag{A.1}$$

Here $O(t^3)$ collects the omitted terms whose magnitude is bounded by a constant times $|t|^3$ as t approaches zero. These terms become smaller than the retained quadratic term. The same notation below describes the size of each omitted remainder near zero.

The fixed margins make the pointwise MI in cell (i, j)

$$\ell_{P_t}(i, j) = \log\left(\frac{q_{ij} + tH_{ij}}{q_{ij}}\right) = \log\left(1 + \frac{tH_{ij}}{q_{ij}}\right) = \frac{tH_{ij}}{q_{ij}} - \frac{t^2 H_{ij}^2}{2q_{ij}^2} + O(t^3), \tag{A.2}$$

using $\log(1+z) = z - z^2/2 + O(z^3)$ with $z = tH_{ij}/q_{ij}$. Multiplying by the cell probability gives

$$\begin{aligned}(q_{ij} + tH_{ij})\ell_{P_t}(i,j) &= (q_{ij} + tH_{ij})\left\{\frac{tH_{ij}}{q_{ij}} - \frac{t^2H_{ij}^2}{2q_{ij}^2} + O(t^3)\right\} \\ &= tH_{ij} - \frac{t^2H_{ij}^2}{2q_{ij}} + \frac{t^2H_{ij}^2}{q_{ij}} + O(t^3) \\ &= tH_{ij} + \frac{t^2H_{ij}^2}{2q_{ij}} + O(t^3).\end{aligned}$$

Multiplication by q_{ij} supplies the negative quadratic term; multiplication by tH_{ij} supplies the positive one. Their combination leaves half the positive term. When we sum over cells, the first-order terms cancel because $\sum_{i,j} H_{ij} = 0$. Thus $f(0) = f'(0) = 0$, $f''(0) = \sum_{i,j} H_{ij}^2/q_{ij}$, and

$$I(P_t) = \frac{t^2}{2}\sum_{i,j}\frac{H_{ij}^2}{q_{ij}} + O(t^3). \tag{A.3}$$

To calculate $V(P_t)$, first square the pointwise-MI expansion:

$$\ell_{P_t}(i,j)^2 = \left\{\frac{tH_{ij}}{q_{ij}} + O(t^2)\right\}^2 = \frac{t^2H_{ij}^2}{q_{ij}^2} + O(t^3).$$

Multiplying by the cell probabilities and summing gives its second moment:

$$\begin{aligned}\mathrm{E}_{P_t}\{\ell_{P_t}(X,Y)^2\} &= \sum_{i,j}(q_{ij} + tH_{ij})\left\{\frac{t^2H_{ij}^2}{q_{ij}^2} + O(t^3)\right\} \\ &= t^2\sum_{i,j}\frac{H_{ij}^2}{q_{ij}} + O(t^3).\end{aligned}$$

MI itself begins with a quadratic term, so $I(P_t)^2 = O(t^4)$. Subtracting it from this second moment leaves the quadratic term unchanged:

$$V(P_t) = t^2\sum_{i,j}\frac{H_{ij}^2}{q_{ij}} + O(t^3). \tag{A.4}$$

The row conditional mean also has no first-order term. Its denominator remains a_i because the margins are fixed, and

$$\begin{aligned}\mathrm{E}_{P_t}\{\ell_{P_t}(X,Y) \mid X=i\} &= \frac{1}{a_i}\sum_j (q_{ij}+tH_{ij})\left\{\frac{tH_{ij}}{q_{ij}}+O(t^2)\right\} \\ &= \frac{t}{a_i}\sum_j H_{ij} + O(t^2) = O(t^2).\end{aligned}$$

The last equality uses the zero row sum of H. The column mean has the same cancellation, using $\sum_i H_{ij} = 0$.

In Equation (4.30), the squared centred pointwise MI, $V(P_t)$, the conditional means and $I(P_t)$ all begin at order t^2. Only the unsquared pointwise-MI term in the bracket has a first-order contribution. Therefore Equation (4.30) supplies the first-order term:

$$g_{P_t}(i,j) = \frac{2tH_{ij}}{q_{ij}} + O(t^2). \tag{A.5}$$

Since g_{P_t} has mean zero, its variance is its second moment:

$$\begin{aligned}\tau^2(P_t) &= \sum_{i,j}(q_{ij}+tH_{ij})\left\{\frac{2tH_{ij}}{q_{ij}}+O(t^2)\right\}^2 \\ &= 4t^2\sum_{i,j}\frac{H_{ij}^2}{q_{ij}} + O(t^3).\end{aligned} \tag{A.6}$$

Comparing the leading terms of $I(P_t)$, $V(P_t)$ and $\tau^2(P_t)$ gives $V(P_t) \approx 2I(P_t)$ and $\tau^2(P_t) \approx 4V(P_t)$ for small positive MI along this path. At $t = 0$, the first-order sampling variance is zero; Appendix A.2 treats that case separately.

A.2 The independence boundary

At independence, $p_{ij} = p_{i+}p_{+j}$, so every population pointwise-MI value is $\ell_P(i,j) = \log 1 = 0$. Consequently, $I(P) = V(P) = 0$, and the first-order change in MI is zero. Sampling can still produce positive estimated MI, but its leading variation comes from second-order cell errors.

To calculate this variation, fix a strictly positive independent table P and let $h = (h_{ij})$ describe a redistribution of probability, with $\sum_{i,j} h_{ij} = 0$. Define

$$P_\varepsilon = P + \varepsilon h, \qquad f(\varepsilon) = I(P_\varepsilon), \tag{A.7}$$

for ε small enough that every cell remains positive. The expansion point is $\varepsilon = 0$, where $f(0) = I(P) = 0$. The second-order Taylor formula gives

$$f(\varepsilon) = f(0) + f'(0)\varepsilon + \frac{f''(0)}{2!}\varepsilon^2 + o(\varepsilon^2). \tag{A.8}$$

The first derivative is

$$f'(0) = \sum_{i,j} \{\ell_P(i,j) - 1\} h_{ij} = -\sum_{i,j} h_{ij} = 0. \tag{A.9}$$

Here the pointwise-MI terms vanish at independence, and the remaining terms cancel because the probabilities must still sum to one.

The row and column changes are $h_{i+} = \sum_j h_{ij}$ and $h_{+j} = \sum_i h_{ij}$. Applying $\mathrm{d}^2(u \log u)/\mathrm{d}u^2 = 1/u$ to the joint and marginal terms in Equation (2.11) gives

$$f''(0) = \sum_{i,j} \frac{h_{ij}^2}{p_{ij}} - \sum_i \frac{h_{i+}^2}{p_{i+}} - \sum_j \frac{h_{+j}^2}{p_{+j}}. \tag{A.10}$$

Thus the leading change is

$$I(P_\varepsilon) = \frac{\varepsilon^2}{2} \left\{ \sum_{i,j} \frac{h_{ij}^2}{p_{ij}} - \sum_i \frac{h_{i+}^2}{p_{i+}} - \sum_j \frac{h_{+j}^2}{p_{+j}} \right\} + o(\varepsilon^2). \tag{A.11}$$

The first sum measures squared cell changes, whereas the last two remove the contributions due to changed margins, leaving the change in association. The remainder is negligible relative to ε^2 as ε approaches zero.

For a multinomial sample, take $\varepsilon h = \widehat{P} - P$. Since cell-probability errors typically have size proportional to $n^{-1/2}$, squaring them makes $\widehat{I}(P)$ proportional to n^{-1} under independence. Under fixed table dimensions and positive cell probabilities, the likelihood-ratio result is

$$2n\widehat{I}(P) \xrightarrow{d} \chi^2_{(r-1)(c-1)} \tag{A.12}$$

(Wilks, 1938; Brillinger, 2004). This is the usual one-table likelihood-ratio, or G, test of independence. Its chi-squared distribution comes from quadratic cell errors, whereas Normal Wald and Expanded Welch use the first-order MI error. The two-sample first-order approximation therefore does not apply unchanged at exact independence.

Sampling variation at small positive MI.

The quadratic approximation also explains why estimated degrees of freedom remain several units when population MI is almost zero. Keep r and c fixed and let a positive population table approach independence with cell departures proportional to $n^{-1/2}$. The normal sampling errors and the population departure then contribute on the same scale. In regular models, the likelihood-ratio statistic has a noncentral chi-squared limit under local alternatives (Banerjee, 2005). For the multinomial model here, assume strictly positive limiting cell probabilities. The local approximation is

$$G = 2n\widehat{I}(P) \mathrel{\dot\sim} \chi^2_d(\kappa), \qquad d = (r-1)(c-1), \qquad \kappa \simeq 2nI(P), \tag{A.13}$$

where $\mathrel{\dot\sim}$ denotes an approximate distribution. The noncentral chi-squared distribution describes a sum of squared normal errors with a nonzero mean departure; κ measures that departure. At independence, $\kappa = 0$, recovering Equation (A.12).

For a sampled table close to independence, the expansions in Appendix A.1, applied around its own independent margins, give

$$\widehat{V}(P) \approx 2\widehat{I}(P) = \frac{G}{n}, \qquad \widehat{\tau}^2(P) \approx 4\widehat{V}(P) \approx \frac{4G}{n}. \tag{A.14}$$

Substitution into the implemented component formula gives

$$\widehat{\nu}_P \approx \frac{2n(G/n)^2}{4G/n} = \frac{G}{2}. \tag{A.15}$$

Thus sampling noise contributes to the estimated degrees of freedom even when the population first-order approximation $nI(P)$ is very small.

The same approximation separates that plug-in calculation from matching the actual sampling moments. Under the noncentral chi-squared approximation, $\mathrm{E}(G) \approx d + \kappa$ and $\mathrm{Var}(G) \approx 2(d + 2\kappa)$, giving

$$\mathrm{E}\{\widehat{V}(P)\} \approx \frac{d+\kappa}{n}, \qquad \mathrm{Var}\{\widehat{V}(P)\} \approx \frac{2(d+2\kappa)}{n^2}. \tag{A.16}$$

Keeping both terms in the moment-matching equation gives

$$\nu_{P,\mathrm{mom}} \approx \frac{2\{(d+\kappa)/n\}^2}{2(d+2\kappa)/n^2} = \frac{(d+\kappa)^2}{d+2\kappa}. \tag{A.17}$$

When κ is much larger than d, this reduces to $\kappa/2 \simeq nI(P)$, as in Equation (4.52). When κ is small, however, the matched degrees of freedom approach d, while the mean plug-in degrees of freedom approach $d/2$.

For the 4×4, $n = 1{,}000$, $I(P) = 0.0001$ setting, $d = 9$ and $\kappa \simeq 0.2$. The local calculation gives a mean estimated component degree of freedom of approximately 4.6, rather than the population first-order value 0.1. It also predicts that the average plug-in estimate of $\mathrm{Var}\{\widehat{V}(P)\}$ divided by its actual sampling variance is approximately

$$\frac{4(d+\kappa)/n^2}{2(d+2\kappa)/n^2} = \frac{2(d+\kappa)}{d+2\kappa} \simeq 1.96. \tag{A.18}$$

This explains the scale of the estimated degrees of freedom in Figure 6.3 and the roughly twofold sample-based variance overprediction discussed alongside it. The population formula alone omits the quadratic sampling contribution responsible for both effects.

Appendix B

Experimental Details

This appendix specifies the population construction, target-MI search, settings underlying the main results, sampling procedure and outcome calculations. Together with the test formulas in Chapters 3–5, these details define the statistical experiment. The final section records the application comparison family used for multiple-testing adjustment.

B.1 Copying construction and target selection

For the copying construction in Equation (5.1), summing a row gives

$$\sum_j p_{ij}(\lambda) = (1-\lambda)a_i \sum_j a_j + \lambda a_i = a_i. \tag{B.1}$$

Summing a column similarly gives a_j, so both margins are preserved. All cells are positive for $0 \leq \lambda < 1$.

MI increases continuously from zero to the entropy of a. With the margins fixed, differentiating the joint terms gives

$$\frac{\mathrm{d}^2 I\{P(\lambda)\}}{\mathrm{d}\lambda^2} = \sum_{i,j} \frac{(a_i \mathbf{1}_{\{i=j\}} - a_i a_j)^2}{p_{ij}(\lambda)} > 0. \tag{B.2}$$

The first derivative at zero is zero, because the change in probabilities sums to zero and all pointwise MI values are zero there. The positive second derivative makes the first derivative positive for $\lambda > 0$. At the

endpoint, $Y = X$, so

$$I\{P(1)\} = -\sum_i a_i \log a_i. \tag{B.3}$$

These facts give a unique copying probability for each positive target below the entropy endpoint.

For each population, calculate this endpoint and solve $I\{P(\lambda)\} - I_{\text{target}} = 0$ on $[0, 1]$ using Brent's method (Brent, 1973). Absolute and relative root tolerances are 10^{-15} and 10^{-14}. The endpoint brackets the search; sampled populations use the interior solution. Accept a table only when every cell is positive, its total and marginal errors are at most 10^{-12}, and its MI error is at most $10^{-12} + 10^{-8} I_{\text{target}}$. Every target is checked against its own population entropy endpoint.

B.2 Parameter grids and sampling

Table 5.2 gives the approximation, broad null and power grids. The power study uses Equation (5.6) for every table size, margin regime and sample size in its grid. The same copying formula and root search apply throughout.

The stronger-common-MI checks in Section 6.2.2 retain the margins and sample sizes of the selected null cases A and B. Their common MI targets are 0.329933611140 nats for A and 0.280551105996 nats for B. Solve separately for the copying probability of each population at its specified target.

For one repetition, draw

$$(N_{P,11}, \ldots, N_{P,kk}) \sim \text{Multinomial}(n_P; p_{11}, \ldots, p_{kk}), \tag{B.4}$$

and independently draw the corresponding Q counts. Reshape these counts into their $k \times k$ tables and apply both tests to those same two sampled tables. Repeat independently 100,000 times per distinct evaluation setting. The simulations use the PCG64 pseudo-random generator, with master seed 20261021 for the approximation and null studies and 20261101 for the fixed-gap power study. Separate streams distinguish settings and the two populations. Repeated settings use one set of repetitions in every relevant display.

All samples retain their declared table dimensions, including empty rows and columns. Use natural logarithms, $0 \log 0 = 0$, and no smoothing. The bias correction for each plug-in MI is $(k-1)^2/(2n)$; do

not truncate negative corrected estimates. In the equal-sample studies, the two bias corrections cancel in their difference. Calculate the standard error using Equation (3.9) and the component and combined degrees of freedom using Chapter 4.

B.3 Validity and sparse-sample measures

Normal Wald requires a finite corrected difference, a finite positive standard error, $\widehat{V}(P)+\widehat{V}(Q) > 10^{-14}$, and a finite p-value. The threshold treats variances extremely close to zero as numerically unusable. Expanded Welch additionally requires positive finite degrees of freedom for both components and their combination, and a finite expanded p-value. A failed component makes Expanded Welch invalid under this rule.

For positive component degrees of freedom, substituting Equation (4.44) and its Q analogue into Equation (4.45) gives the equivalent combined form

$$\widehat{\nu}_{\text{expanded}} = \frac{\left\{\widehat{V}(P)/n_P + \widehat{V}(Q)/n_Q\right\}^2}{\widehat{\tau}^2(P)/(2n_P^3) + \widehat{\tau}^2(Q)/(2n_Q^3)}. \tag{B.5}$$

This direct form remains defined when one table has $\widehat{V} = \widehat{\tau}^2 = 0$ and the other gives a positive denominator. The zero contribution drops out, leaving the other table's component degrees of freedom. For the zero component, however, Equation (4.44) requires $0/0$. The specified componentwise algorithm requires both component degrees of freedom to be positive and finite, so it records this case as invalid even though the direct combined expression remains defined.

The primary rate divides valid rejections by all repetitions. A method-valid rate uses only that method's valid repetitions; a common-valid rate uses only repetitions where both tests are valid. Report the denominator for either conditional rate, which is undefined when the denominator is zero. Invalid outputs contribute no rejection to the primary rate.

Record the number of zero-count cells, empty rows and empty columns separately for each population. These are sampling diagnostics rather than eligibility rules. The minimum expected cell count is $\min_{ij} np_{ij}$, with the corresponding definition for Q. The study applies no minimum expected-count filter.

B.4 Bias, variance and degrees of freedom

Let $\widehat{\Delta}_s$ be the corrected difference in repetition s. Its bias estimate is the mean of the finite-valued MI difference estimates minus the known Δ. Its observed sampling variance is

$$\frac{1}{M-1}\sum_{s=1}^{M}\left(\widehat{\Delta}_s - \frac{1}{M}\sum_{u=1}^{M}\widehat{\Delta}_u\right)^2, \tag{B.6}$$

where M counts the finite-valued differences, including those whose test output is invalid. Centring at their sample mean separates variance from bias. For P and Q, the individual corrected MI biases are the mean finite corrected MI estimates minus $I(P)$ and $I(Q)$, respectively.

For the estimated / observed variance ratio, restrict the calculations to the same repetitions with a finite difference and finite squared standard error. Retain finite zero variance estimates. Divide the mean squared standard error by the observed variance on this matched subset. Report the included fraction. The ratio is defined only when the observed variance is positive. The population comparison instead uses $V(P)/n_P + V(Q)/n_Q$ and the primary observed variance.

The Monte Carlo standard error of estimated bias is the observed SD divided by $\sqrt{M}$. Component and combined degrees-of-freedom summaries retain positive finite values and report their included fractions. Finite zero variance estimates remain in moment calculations even when component degrees of freedom are undefined. Figures divide the mean estimate by its moment target only where every component estimate is defined; conditional summaries elsewhere retain their inclusion counts. These diagnostics do not isolate each estimation error's causal contribution to rejection.

Sampling variability of the estimated pointwise-MI variance.

For each selected population, measure the observed variability of $\widehat{V}(P)$ by replacing $\widehat{\Delta}_s$ with $\widehat{V}_s(P)$ in Equation (B.6). This measures how much the variance estimate itself changes between samples. Chapter 4 predicts that variability to first order. Compare the observed value with

$$\begin{aligned}\text{population prediction} &= \frac{\tau^2(P)}{n_P},\\ \text{mean plug-in prediction} &= \frac{1}{M}\sum_{s=1}^{M}\frac{\widehat{\tau}_s^2(P)}{n_P}.\end{aligned} \tag{B.7}$$

The population prediction uses the known probabilities, whereas the mean plug-in prediction uses the empirical probabilities in each repetition. Repeat both for Q. For each population, use the same repetitions with finite $\widehat{V}_s$ and $\widehat{\tau}_s^2$, including finite zeros and outputs whose test is invalid. Divide both predictions by the same observed variability. A ratio below one indicates underprediction, a ratio above one indicates overprediction, and one indicates agreement. The observed variability must be positive; a zero prediction gives a zero ratio.

B.5 Simulation uncertainty

For a rate estimate $\widehat{r}$ from M independent repetitions, its approximate Monte Carlo standard error is $\{\widehat{r}(1-\widehat{r})/M\}^{1/2}$. Rejection and validity rates use pointwise 95% Wilson intervals (Wilson, 1927).

For the paired method contrast, let D_s equal the Expanded Welch valid-rejection indicator minus Normal Wald's valid-rejection indicator in repetition s. The approximate interval for the average paired difference $\bar{D}$ is

$$\bar{D} \pm 1.96 \left\{ \frac{1}{M(M-1)} \sum_{s=1}^{M} (D_s - \bar{D})^2 \right\}^{1/2}. \tag{B.8}$$

Use the same common-valid subset when reporting a conditional paired contrast. All intervals describe simulation uncertainty at fixed populations, not simultaneous coverage over every setting.

B.6 Component moment targets and uncertainty

The population first-order degrees of freedom $\nu_{P,\mathrm{FO}}$ in Equation (4.40) approximate the sampling-moment target $\nu_{P,\mathrm{mom}}$ in Equation (4.8). The target uses the actual sampling mean and variance of $\widehat{V}(P)$, measured over repeated samples from the same population at fixed sample size. Assuming positive $V(P)$, $\tau^2(P)$ and sampling variance, divide the two definitions and rearrange to obtain the exact identity

$$\frac{\nu_{P,\mathrm{FO}}}{\nu_{P,\mathrm{mom}}} = \frac{\left\{V(P)/\,\mathrm{E}\,\widehat{V}(P)\right\}^2}{\{\tau^2(P)/n_P\}/\,\mathrm{Var}\{\widehat{V}(P)\}}. \tag{B.9}$$

The numerator is the reciprocal of the squared mean ratio in Figure 6.1; the denominator is the middle-row predicted / observed sampling-variance ratio. A sampling mean $\mathrm{E}\{\widehat{V}(P)\}$ above $V(P)$ lowers the

first-order degrees of freedom relative to the moment target when the variance prediction is otherwise accurate. The same calculations are made separately for Q.

Estimating the targets and their uncertainty.

For a component, let $X_s = \widehat{V}_s$ in repetition s. Use all finite values, including zeros, even if component degrees of freedom are undefined. From M repetitions, calculate

$$\bar{X} = \frac{1}{M}\sum_{s=1}^{M} X_s, \qquad S_X^2 = \frac{1}{M-1}\sum_{s=1}^{M}(X_s - \bar{X})^2. \tag{B.10}$$

For $S_X^2 > 0$, the estimated moment target is $2\bar{X}^2/S_X^2$. The diagnostic decomposition uses these evaluation moments.

The delta method uses a first-order Taylor expansion of $f(\mu, v) = 2\mu^2/v$, where μ and v are the sampling mean and variance of $\widehat{V}(P)$. Its derivatives are $\partial f/\partial\mu = 4\mu/v$ and $\partial f/\partial v = -2\mu^2/v^2$. Evaluating them at $\bar{X}, S_X^2$ gives the two coefficients below. The mean and variance errors are propagated jointly because both estimates use the same repetitions. Define

$$U_s = X_s - \bar{X}, \qquad W_s = (X_s - \bar{X})^2 - S_X^2. \tag{B.11}$$

The first-order contribution to the target's estimation error is

$$Z_s = \frac{4\bar{X}}{S_X^2}U_s - \frac{2\bar{X}^2}{(S_X^2)^2}W_s. \tag{B.12}$$

Its sample SD divided by $\sqrt{M}$ estimates the Monte Carlo standard error. Calculating the SD after combining the two terms retains the mean–variance covariance.

Mean and variance ratios use the same joint delta-method calculation, keeping the plug-in prediction and observed variance paired.

B.7 Application comparison family and Holm adjustment

The shopping comparisons in Section 6.4 were selected from the six comparisons in Table B.1. For Adult, the comparisons use education or occupation versus an indicator of annual income above $50,000,

comparing female and male records (Becker and Kohavi, 1996). For Student Performance, they use study time or maternal education versus an indicator of final grade at least 10 in the Portuguese course, comparing records from schools labelled GP and MS (Cortez, 2008). The shopping variables and visitor groups are defined in Section 6.4.

Table B.1: Unadjusted p-values in the six-comparison application family.

Dataset	Predictor	Normal Wald	Expanded Welch
Adult	Occupation	1.184×10^{-44}	1.635×10^{-42}
Adult	Education	1.890×10^{-34}	4.930×10^{-33}
Online Shoppers	Month	4.714×10^{-5}	3.726×10^{-4}
Student Performance	Study time	0.1982	0.2339
Student Performance	Maternal education	0.8361	0.8468
Online Shoppers	Weekend	0.9554	0.9601

Holm's adjustment is applied separately to each method's six p-values (Holm, 1979). Order them from smallest to largest and multiply them by 6, 5, 4, 3, 2 and 1, respectively, then take the cumulative maximum and cap the adjusted values at one. This calculation gives the adjusted month p-values reported in Section 6.4.

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